

Autodesk® **Scaleform**®

Scaleform CLIK AS3

Scaleform CLIK

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2.0

2010 8 19

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Autodesk® Scaleform® 4.2

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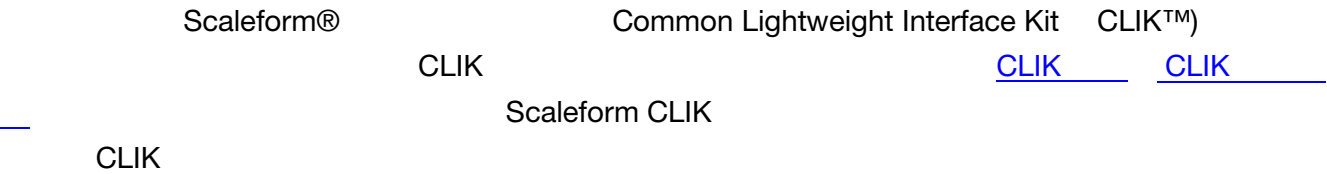
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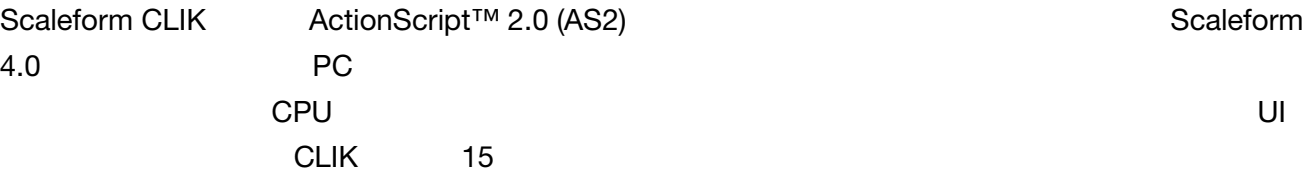
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1



1.1



1.1.1. Scaleform CLIK

UI

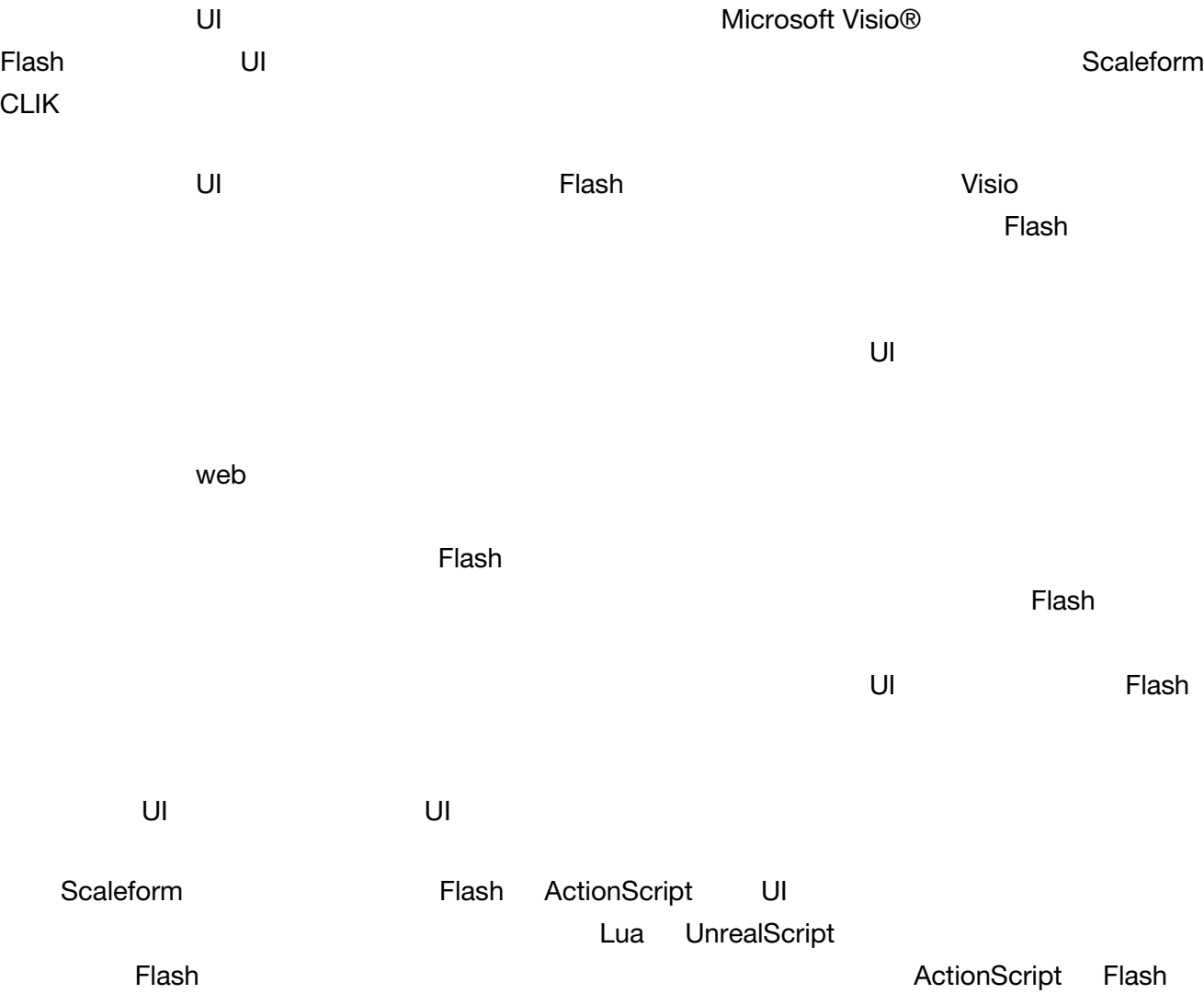
Button	Slider
ButtonBar	StatusIndicator
CheckBox	TileList
RadioButton	Label
TextInput	ScrollingList
TextArea	DropDownMenu
ScrollIndicator	NumericStepper
ScrollBar	

API

InputManager	UIComponent
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
IUIComponent	IList
Constraints	IListItemRenderer

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK AS3 API Reference
CLIK AS3 User Guide
CLIK Flash Samples
CLIK Video Tutorials

1.2 UI



1.3

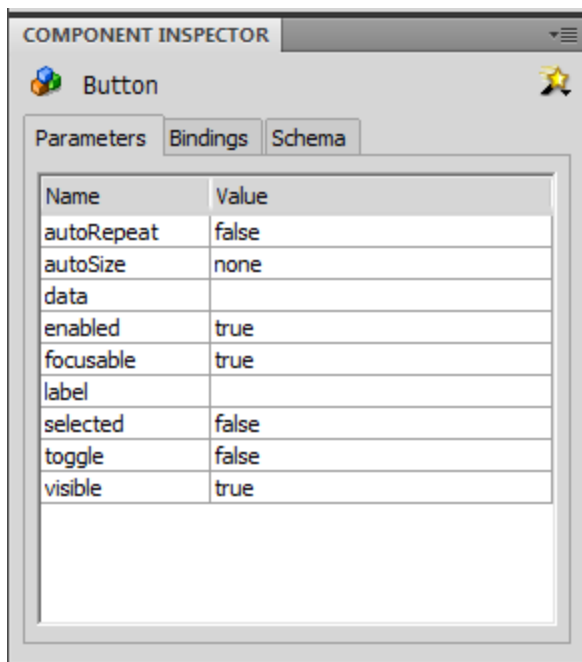
Flash
Flash
Scaleform CLIK
Scaleform
gskinner.com

gskinner.com Grant Skinner Grant Skinner Adobe Flash Creative
Suite® 4 (CS4) Flash gskinner.com
<http://gskinner.com/blog>

CLIK Flash studio CLIK
Scaleform SDK\Resources\AS3\CLIK\components\CLIK_Components_AS3 fla

1.3.1 屬性檢查

Flash IDE Component Inspector Parameters
CS4 Window Component Inspector
(Shift+F7)
AS



CLIK

SWF

Flash IDE

1.4 框架基礎知識

1.4.1 事件

CLIK

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ActionScript 3 Event , Adobe ActionScript 3
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

Scaleform , Scaleform , Extensions.enabled
true() ,Scaleform MouseEventEx, MouseEvent (MouseEvent)
MouseEventEx / MouseEventEx , ,

1.4.2

Flash , ,stage
InteractiveObject

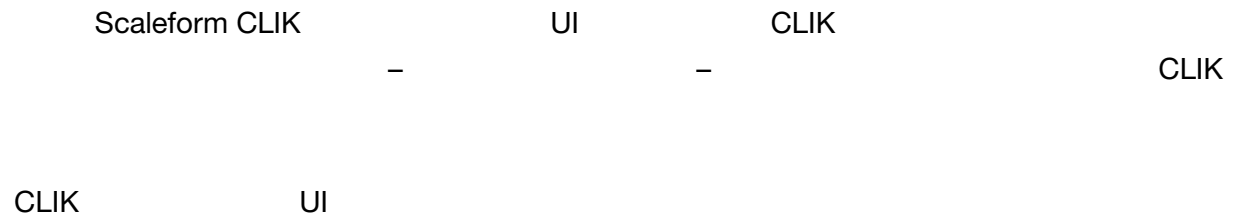
Programming

CLIK CLIK
(Tab) (Shift+Tab)
CLIK CLIK CLIK
Flash CLIK CLIK
(Tab) (Shift+Tab)

Enter

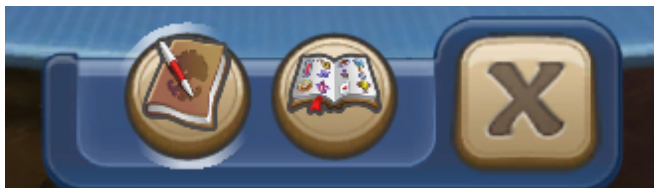
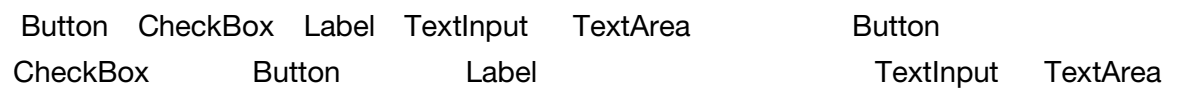
Xbox360 PS3 A X UI Enter
CLIK

2

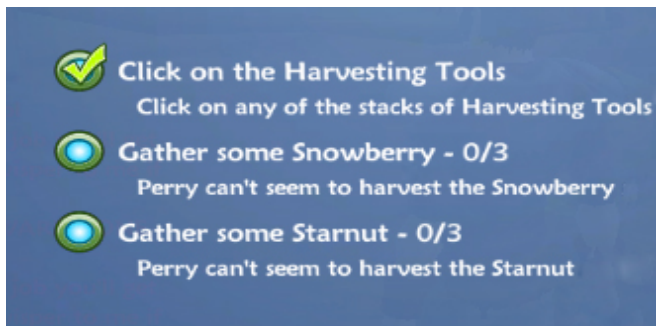


- **User interaction**
- **Component setup** Flash
- **States**
- **Inspectable properties** Flash
- **Events** UI
- **Tips and tricks**

2.1



1 **Free Realms**



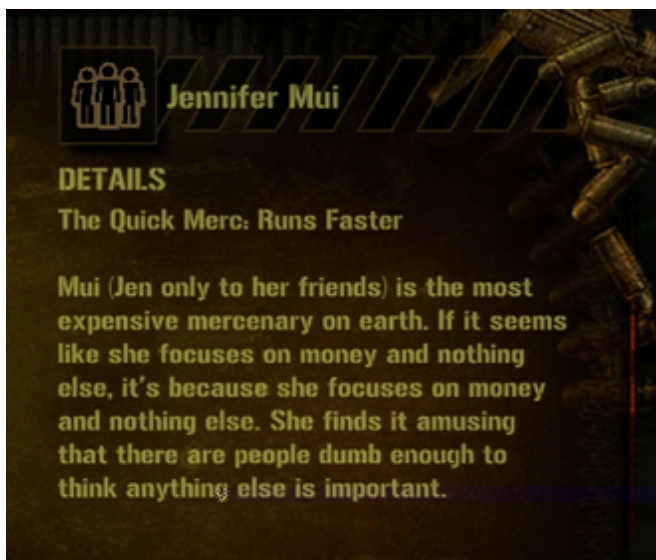
2 *Free Realms* Check box



3 *Free Realms* Label



4: *Crysis Warhead* Text input



5 *Mercenaries 2* Text area

2.1.1 Button



6

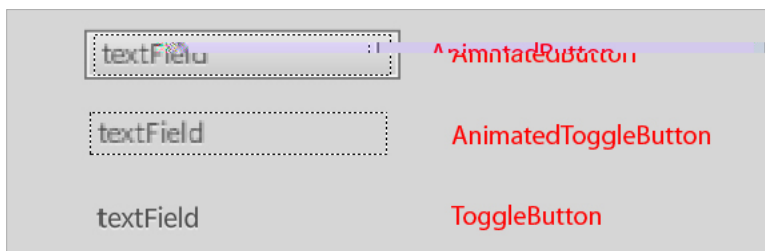
Button CLIK

Button scaleform.clik.controls.Button

textField

ScrollBar

Slider



7 AnimatedButton AnimatedToggleButton ToggleButton

CLIK Button

ToggleButton AnimatedButton AnimatedToggleButton

Button fla

2.1.1.1

2.1.1.2

CLIK Button MovieClip

- **textField:** () TextField
- **focusIndicator:** () MovieClip
show hide MovieClip over

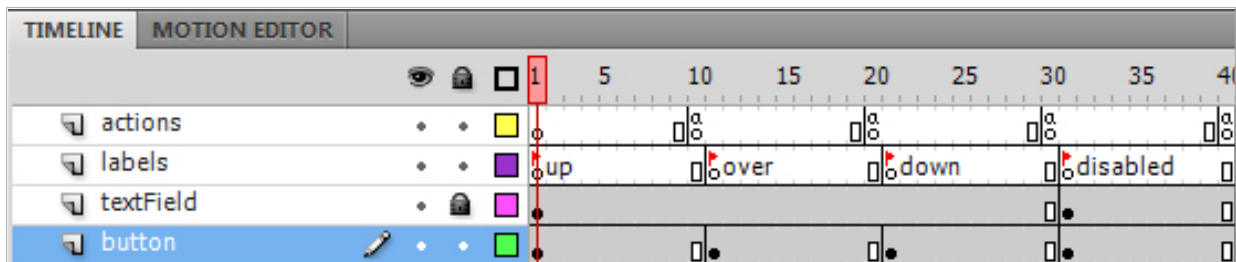
Button

over

2.1.1.3

CLIK

- *up*
- *over*
- *down*
- *disabled*



8 Button

Flash

[CLIK](#)

2.1.1.4

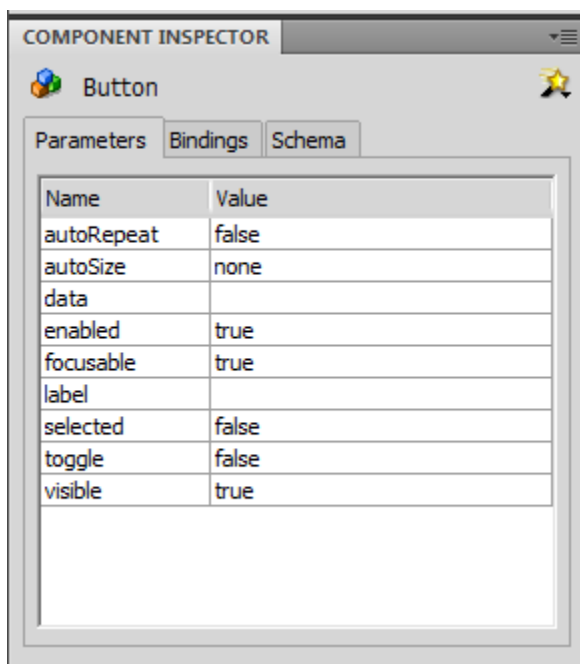


Figure 1: Button component inspectable properties in the CS4 component inspector.

:

autoRepeat	"click" ()
autoSize	autoSize TextFieldAutoSize.NONE
Data	ButtonGroup
enabled	false (),
focusable	/ focusable () false tab
Label	
selected	: toggle () true , , , toggle false, ActionScript
Toggle	true
Visible	false

2.1.1.5

CLIK

Event () Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
MouseEvent.ROLL_OVER	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
MouseEvent.ROLL_OUT	The mouse cursor has rolled out of the button. mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.PRESS	DropDownMenu() controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false – isRepeat: true – autoRepeating ; false –
MouseEvent.DOUBLE_CLICK	doubleClickEnabled true mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/

```
(Gamepad)      ; false ad  
isRepeat: true ea  
autoRepeating   ; false pe
```

2.1.2.1

CheckBox

2.1.2.2

CLIK CheckBox

MovieClip

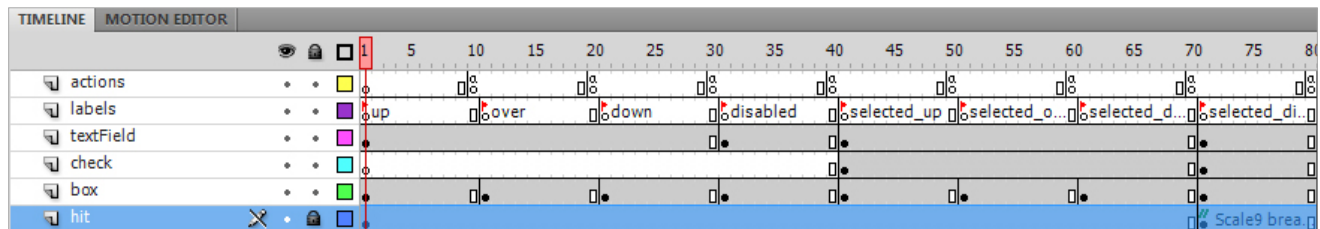
- **textField:** () TextField
 - **focusIndicator:** () MovieClip
- MovieClip show hide

2.1.2.3

Toggle

CheckBox

- **up**
- **over**
- **down**
- **disabled**
- **selected_up**
- **selected_over**
- **selected_down**
- **selected_disabled**



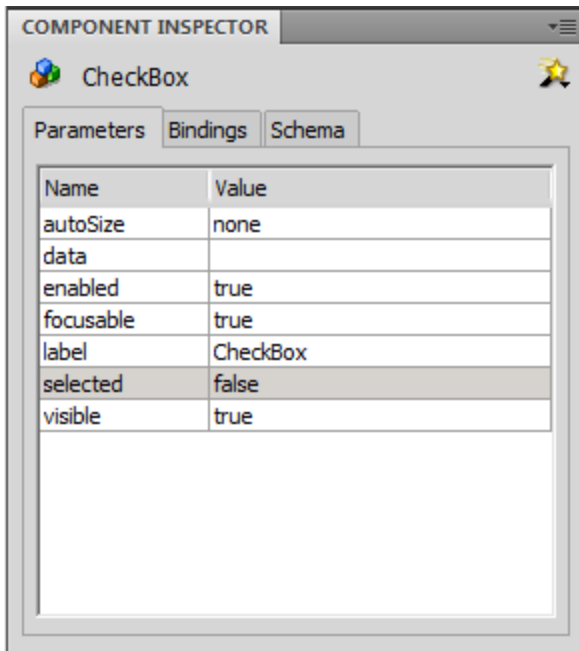
11: CheckBox

CheckBox

Button

CLIK

2.1.2.4



Toggle

CheckBox

autoRepeat

autoSize	
	autoSize
	autoSize=TextFieldAutoSize.NONE
Data	ButtonGroup
enabled	false,
focusable	/ focusable()
	false tab
Label	
selected	:
	toggle() true ,
	, , toggle false,
	ActionScript
Visible	false

2.1.2.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

CheckBox

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
ComponentEvent.STATE_CHANGE	
MouseEvent.ROLL_OVER	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
MouseEvent.ROLL_OUT	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.PRESS	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -

MouseEvent.DOUBLE_CLICK	<i>doubleClickEnabled</i>	true
	mouseIdx:	(
	) uint	Scaleform,
	MouseEventEx	buttonIdx:
	(	Scaleform,
	MouseEventEx	
ButtonEvent.CLICK	controllerIdx:	(
	) uint	
	isKeyboard: true -	(Keyboard)/
	(Gamepad) ; false -	
	isRepeat: true -	
	autoRepeating ; false -	
ButtonEvent.DRAG_OVER	controllerIdx:	(
	) uint	
ButtonEvent.DRAG_OUT	controllerIdx:	(
	) uint	
ButtonEvent.RELEASE_OUTSIDE	controllerIdx:	(
	) uint	

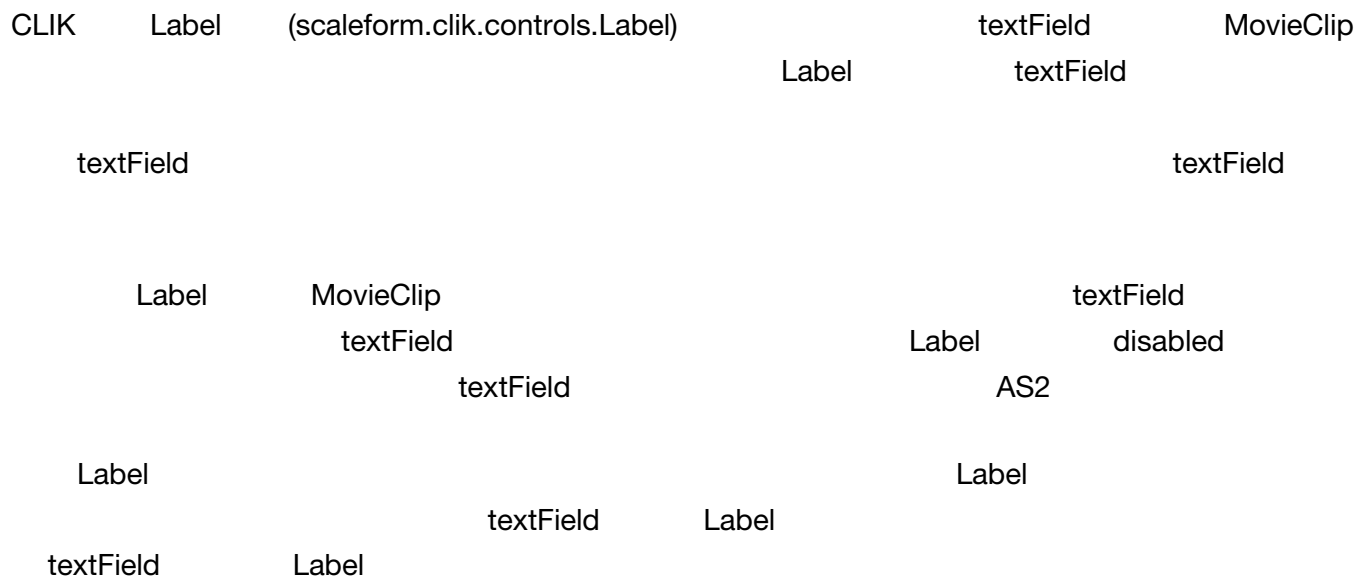
CheckBox Toggle

```
myCheckBox.addEventListener(Event.SELECT, onCheckBoxToggle);
function onCheckBoxToggle(e:Event):void {
    // Do something
}
```

2.1.3 Label



12:



2.1.3.1

Label

2.1.3.2

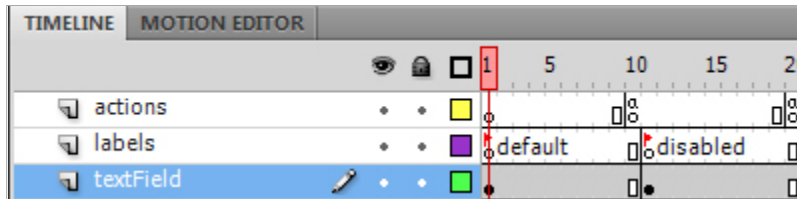
CLIK Label MovieClip

- **textField:** TextField Label

2.1.3.3

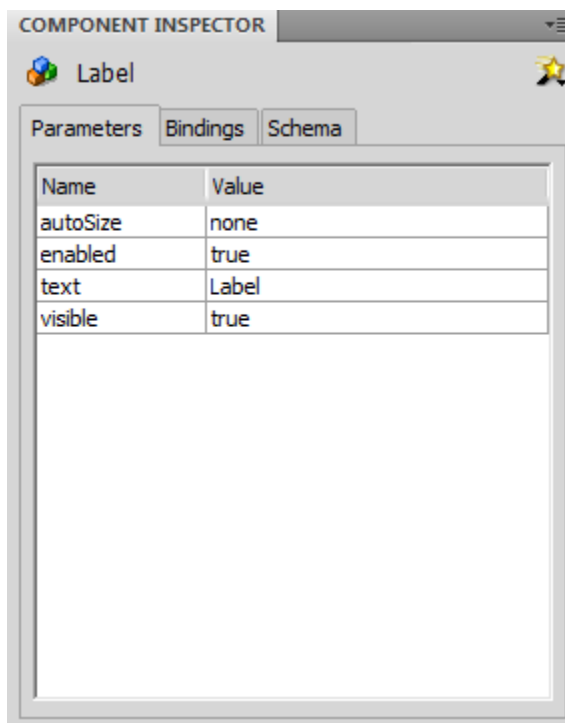
CLIK Label

- **default** enabled
- **disabled**



13: Label

2.1.3.4



Label

Text	
Visible	false,
Enabled	false,
autoSize	(Label)
autoSize	
autoSize=TextFieldAutoSize.NONE	

2.1.3.5

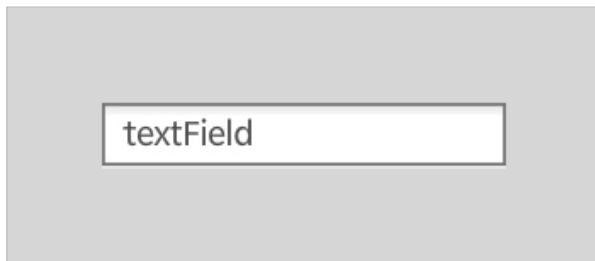
Event() Event ,

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

Label

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		

2.1.4 TextInput



14: TextInput.

TextInput (scaleform.clik.controls.TextInput)

textField

Label

textField

textField

HTML

TextInput textField

textField

Label

Label

textFields

TextInput

UI

PC

TextInput

textField

TextInput disabled textField
 TextInput AS2
 TextInput TextInput
 Flash Studio TextInput
 TextInput textField,
 (Shift+Insert) (Shift+Arrows) (Shift+Delete) (Ctrl+Insert) HTML
 “ ” actAsButton
 “over” “out”
 actAsButton “over”/”out” “ ” “ ”
 “ ”
 (0xAFFFFFFF) “ ” “ ”
 ”

2.1.4.1

TextInput textField
 textField

2.1.4.2

CLIK TextInput MovieClip

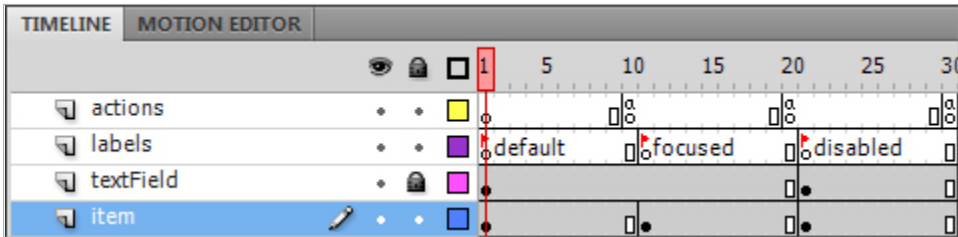
- **textField**: textField

2.1.4.3 States

CLIK TextInput disabled

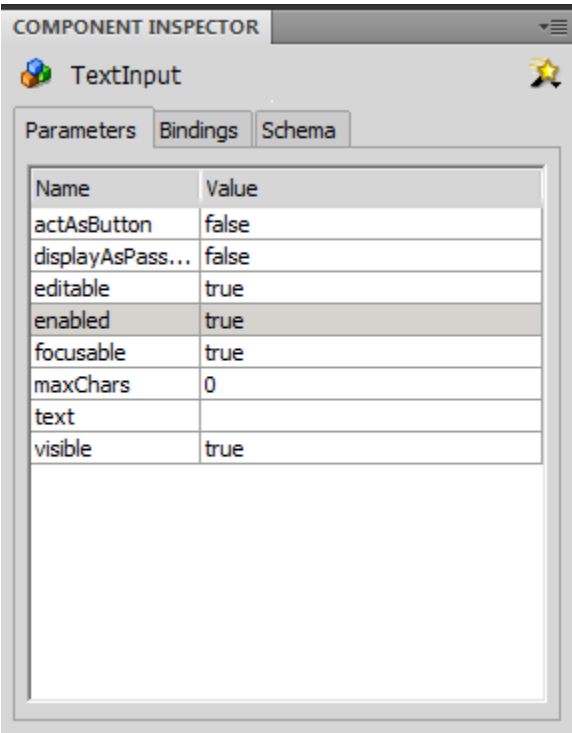
- **default** enabled

- *focused* textField
- *disabled*



15: TextInput

2.1.4.4



TextInput :

actAsButton	“ ” “ ”	tab
displayAsPassword	true, textField “*” , textField ,	
Editable	false, TextInput	
Enabled	Disables the component if set to false.	
Focusable	/	focusable()

	false	tab
maxChars		textField
Text	textField	
Visible	false	

2.1.4.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

TextInput

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.CHANGE	textField	
ComponentEvent.STATE_CHANGE		
MouseEvent.ROLL_OVER	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
MouseEvent.ROLL_OUT	mouseIdx: () uint Scaleform,	

2.1.5.1

TextArea

textField

textField

2.1.5.2

CLIK TextArea

MovieClip

- **textField:** TextField

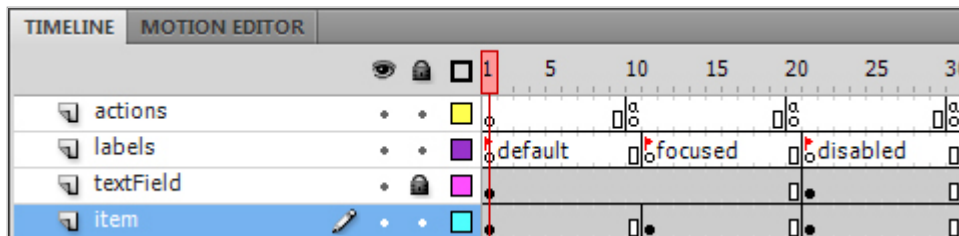
2.1.5.3

TextInput

TextArea

disabled

- **default** enabled
- **focused** textField
- **disabled**



17: TextArea

2.1.5.4

TextArea	TextInput	CLIK
ScrollBar	2.4	
actAsButton		, ,TextInput
displayAsPassword		

Visible	false
---------	-------

2.1.5.5

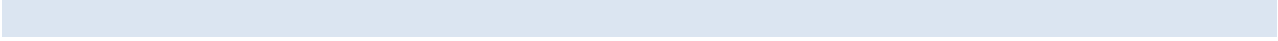
Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable*

TextArea

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.CHANGE	textField	
Event.SCROLL		
ComponentEvent.STATE_CHANGE		
MouseEvent.ROLL_OVER	mouseIdx: () uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
MouseEvent.ROLL_OUT	mouseIdx: () uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx	

TextArea



CLIK CheckBox ButtonGroup

CheckBox
ButtonGroup

ButtonGroup

2.2.1.1

2.2.1.2

CLIK RadioButton MovieClip

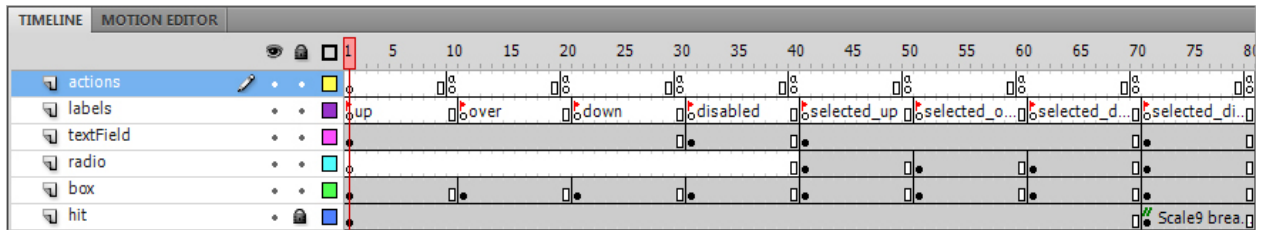
MovieClip

- ***textField*** () TextField
- ***focusIndicator*** MovieClip
MovieClip show hide

2.2.1.3

CheckBox

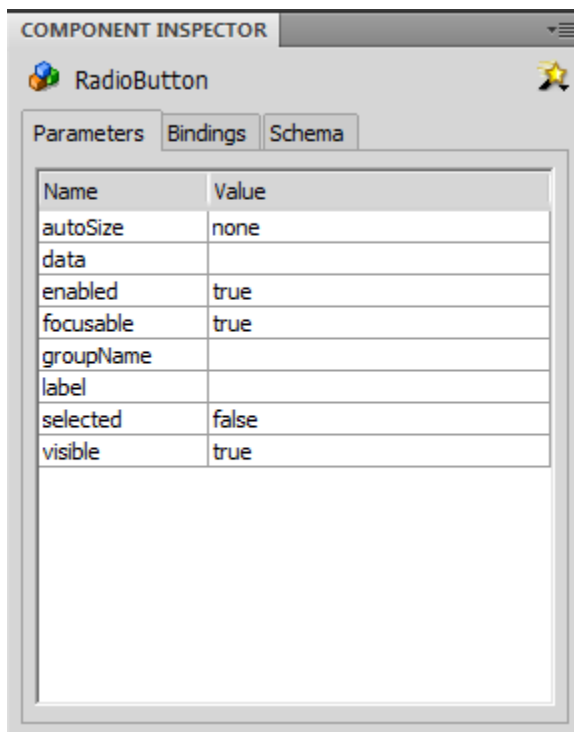
- *up*
-
- *over*
- *down*
- *disabled*
- *selected_up*
-
- *selected_over*
- *selected_down*
- *selected_disabled*



20: RadioButton

RadioButton Button
 RadioButton [CLICK](#)

2.2.1.4



Button RadioButton
 disableFocus Toggle

autoSize

autoSize
 autoSize=TextFieldAutoSize.NONE

Data

ButtonGroup

enabled	false,
focusable	/ focusable() false
	tab
groupName	ButtonGroup , , RadioButton RadioButton , ButtonGroup RadioButton , RadioButton _root , _root ButtonGroup RadioButton
Label	
selected	:
	toggle() true , , , toggle false, ActionScript

2.2.1.5

	Event() Event ,
• type • target • currentTarget Event • eventPhase EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE • bubbles • cancelable	
	RadioButton

ComponentEvent.SHOW	visible() true
ComponentEvent.HIDE	visible false
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
ComponentEvent.STATE_CHANGE	
MouseEvent.ROLL_OVER	

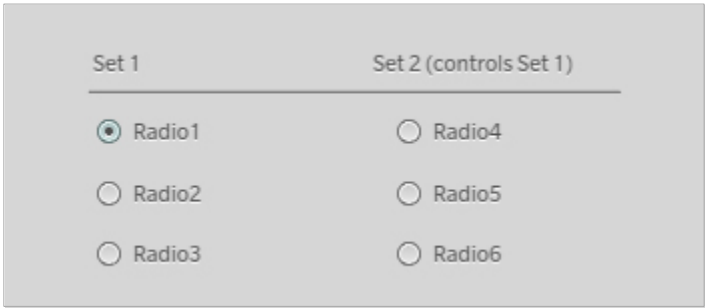
	mousetdx: () uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx
MouseEvent.ROLL_OUT	mousetdx: () uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.PRESS	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -
MouseEvent.DOUBLE_CLICK	<i>doubleClickEnabled</i> true mousetdx: () uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -
ButtonEvent.DRAG_OVER	

	controllerIdx:	(
	) uint	
ButtonEvent.DRAG_OUT		
	controllerIdx:	(
	) uint	
ButtonEvent.RELEASE_OUTSIDE		
	controllerIdx:	(
	) uint	

RadioButton

```
myRadio.addEventListener(Event.SELECT, onRadioToggle);
function onRadioToggle(e:Event):void {
    // Do something
}
```

2.2.2 ButtonGroup



21: ButtonGroup.

CLIK ButtonGroup (scaleform.clik.controls.ButtonGroup)

RadioButton

CLIK
ButtonGroup

CheckBox

2.2.2.1

RadioButton

2.2.2.2

CLIK ButtonGroup MovieClip

2.2.2.3

ButtonGroup

2.2.2.4

ButtonGroup

2.2.2.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ButtonGroup

Event.CHANGE

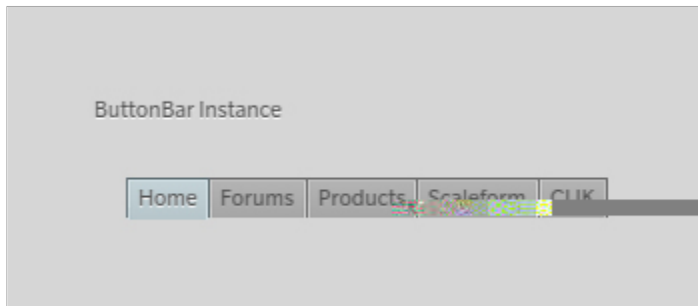
ButtonEvent.CLICK

target: CLIK Button

ButtonGroup

```
myGroup.addEventListener(Event.CHANGE, onNewSelection);
function onNewSelection(e:Event):void {
    if (e.item == myRadio) {
        // Do something
    }
}
```

2.2.3 ButtonBar



22: ButtonBar.

ButtonBar (scaleform.clik.controls.ButtonBar)

dataProvider

tab UI

ButtonGroup

dataProvider

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

2.2.3.1

ButtonGroup

2.2.3.2

CLIK

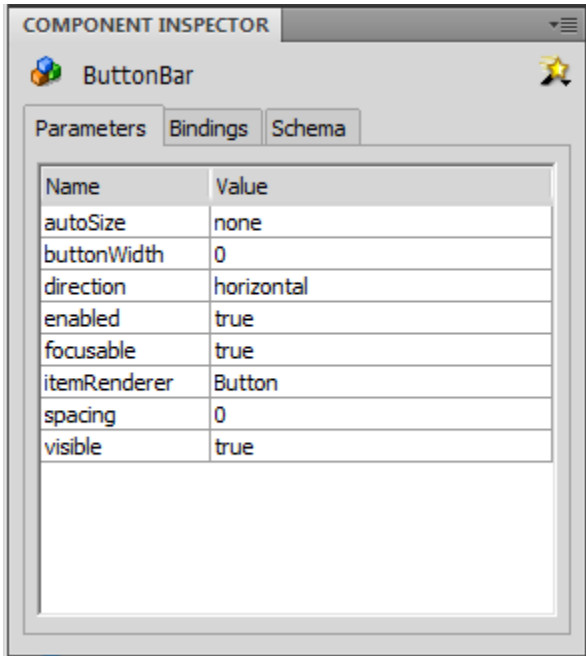
ButtonBar

MovieClip

2.2.3.3

CLIK

2.2.3.4



Flash IDE

ButtonBar

autoSize	true,	Button()
Direction	“(Horizontal)”	Button, “”
	(Vertical)	
buttonWidth	Button	autoSize true,
Enabled	false,	
Focusable	/	focusable() false
	tab	
itemRenderer	ID	
Spacing		direction
Visible	false,	ButtonBar

2.2.3.5

Event() Event ,

- type

- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ButtonBar

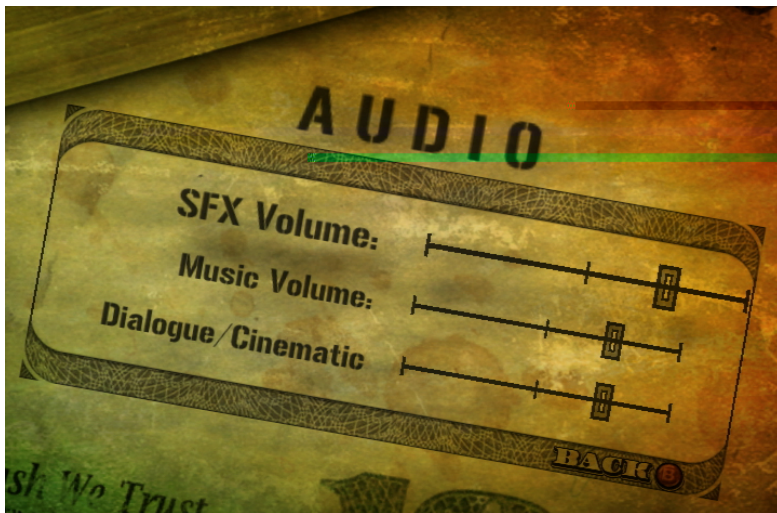
ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
IndexEvent.INDEX_CHANGE	index:ButtonBar) int) 1 data: dataProvider	int -1(1 lastIndex:ButtonBar -1(AS3 Object
ButtonBarEvent.BUTTON_SELECT	index:ButtonBar) renderer:ButotnBar	Int -1(1 Button Button

```
myBar.addEventListener(ButtonBarEvent.BUTTON_SELECT, onItemClick);
function onItemClick(e:ButtonBarEvent) {
    processData(e.renderer.data);
    // Do something
}
```

2.3

```
ScrollIndicator ScrollBar Slider
                ScrollBar
                        Slider
```

ScrollIndicator



23: *Mercenaries 2*

2.3.1 ScrollIndicator

24: ScrollIndicator.

CLIK ScrollIndicator (scaleform.clik.controls.ScrollIndicator)
textField

TextArea	ID
----------	----

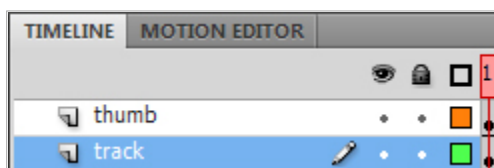
2.3.1.1

2.3.1.2

CLIK ScrollIndicator MovieClip

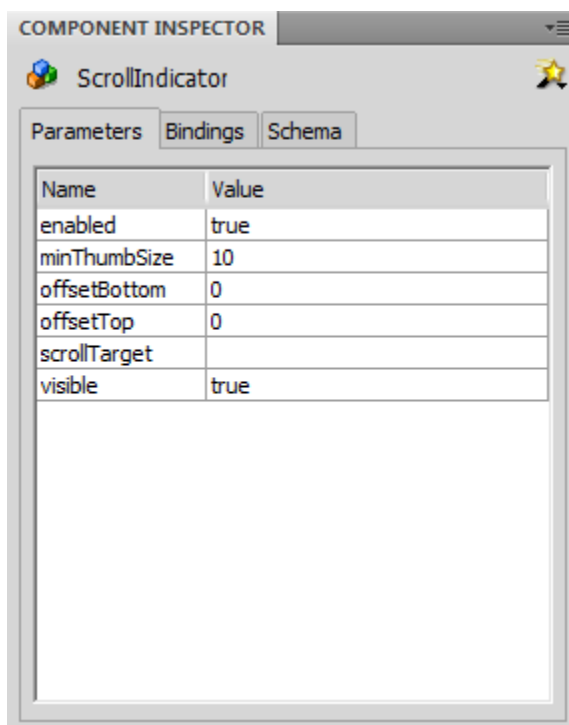
- **thumb**: CLIK Button
- **track** MovieClip

2.3.1.3



25: ScrollIndicator

2.3.1.4



Enabled	false,
offsetTop	
offsetBottom	
scrollTarget	TextArea textField textField ScrollIndicator
Visible	false

2.3.1.5

Event() Event ,

- **type**
-
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ScrollIndicator

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
Event.SCROLL	ScrollIndicator	

```
mySl.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) {
    trace("mySl.position: " + e.target.position);
    // Do something
}
```

2.3.2 ScrollBar



26: ScrollBar.

CLIK ScrollBar(scaleform.clik.controls.ScrollBar)

2.3.2.1

ScrollBar

trackMode
(Shift)

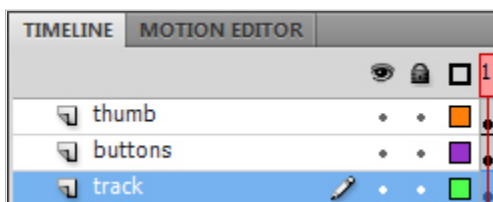
trackMode

2.3.2.2

CLIK ScrollBar MovieClip

- ***upArrow*** CLIK
- ***downArrow***: CLIK
- ***thumb***: CLIK
- ***track***: CLIK

2.3.2.3

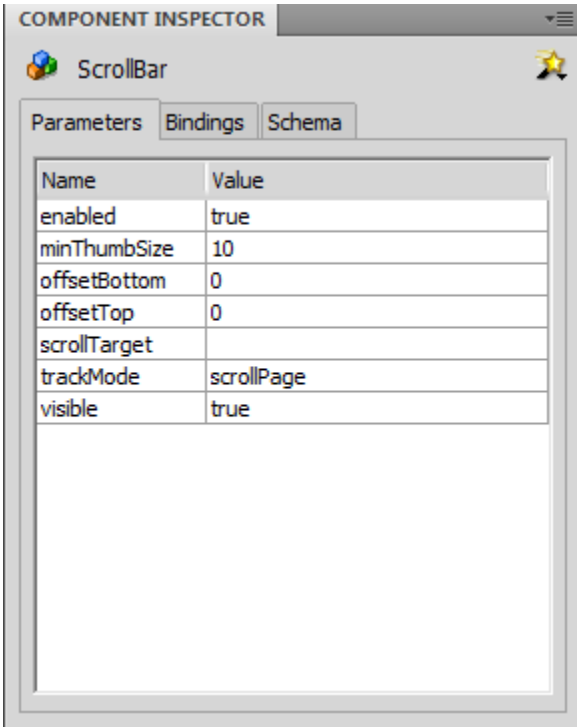


27: ScrollBar

ScrollBar

ScrollIndicator

2.3.2.4



, trackMode:

Enabled	false,
offsetTop	
offsetBottom	
scrollTarget	TextArea textField
trackMode	scrollPage scrollToCursor
Visible	false

2.3.2.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event

- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ScrollBar

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
Event.SCROLL	ScrollBar	

```
mySB.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) {
    trace("mySB.position: " + e.target.position);
    // Do something
}
```

2.3.3 Slider



28: Slider.

Slider(scaleform.clik.controls.Slider)

2.3.3.1

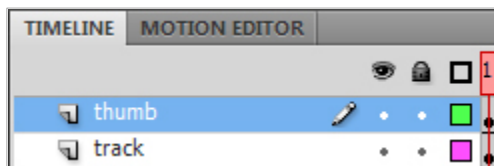
home end

2.3.3.2

CLIK Slider MovieClip

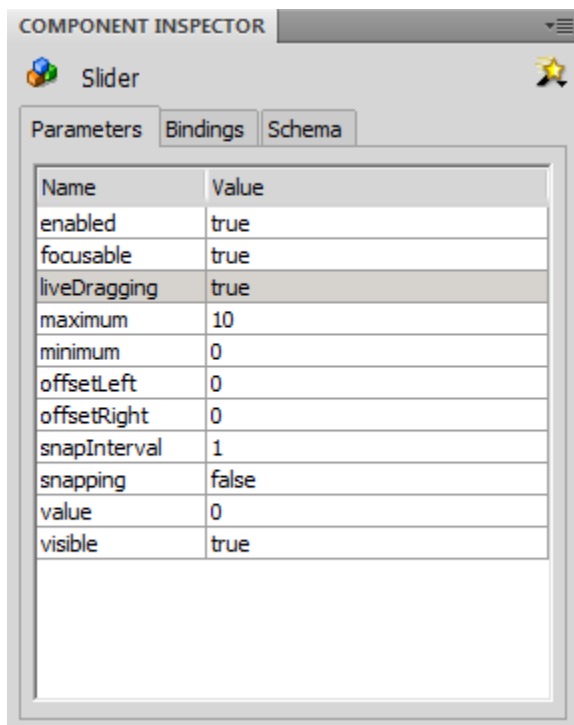
- ***thumb*** CLIK
- ***Track*** CLIK

2.3.3.3



29: Slider

2.3.3.4



Slider

Enabled

false,

Focusable	/	focusable()	false
	tab		
Value	Slider		
Minimum			
Maximum			
Snapping	true		
snapInterval		false	
liveDragging	true	Slider	
offsetLeft			
offsetRight			
Visible	false		

2.3.3.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

Slider

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
SliderEvent.VALUE_CHANGE	Slider's	

Slider

```
mySlider.addListener(SliderEvent.VALUE_CHANGE, onValueChange);
function onValueChange(e:SliderEvent) {
    trace("mySlider.value: " + e.target.value);
    // Do something
}
```

```
}
```

2.4

CLIK NumericStepper OptionStepper ScrollingList TileList DropdownMenu
 NumericStepper DataProvider ListItemRenderer
 ScrollingList TileList

NumericStepper OptionStepper ScrollingList TileList
 ScrollIndicator ScrollBar DropdownMenu
 ScrollingList TileList



30: *Mercenaries 2*

2.4.1.2

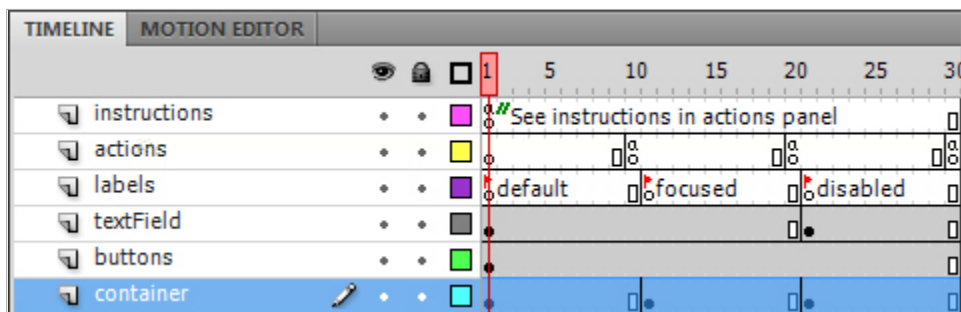
CLIK NumericStepper MovieClip

- **textField**: TextField
- **nextBtn**: CLIK Button
- **prevBtn**: CLIK Button

2.4.1.3

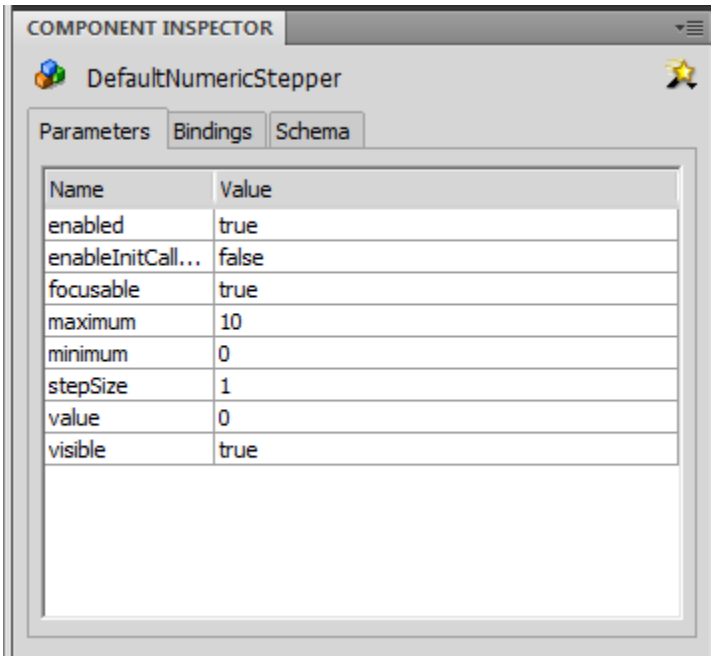
NumericStepper

- **default** enabled
- **focused** textField
- **disabled**



34: NumericStepper

2.4.1.4



NumericStepper

MovieClip

Enabled	false,
Focusable	/ focusable() false
	tab
minimum	NumericStepper
maximum	NumericStepper
stepSize	NumericStepper , / NumericStepper
Value	Slider
Visible	false

2.4.1.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE

- *bubbles*
- *cancelable*

NumericStepper

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
IndexEvent.INDEX_CHANGE	NumericStepper index:NumericStepper 1() lastIndex:NumericStepper -1() 1 data: dataProvider	int 1 int AS3 Object

NumericStepper

```
myNS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
    trace("myNS.value: " + e.target.value);
    // Do something
}
```

2.4.2 OptionStepper



35: **OptionStepper.**

OptionStepper (scaleform.clidik.controls.OptionStepper), NumericStepper
dataProvider

OptionStepper

dataProvider

```
optionStepper.dataProvider = new DataProvider(["item1", "item2", "item3",  
"item4"]);
```

2.4.2.1

NumericStepper

OptionStepper

(Home) (End)

dataProvider

2.4.2.2

CLIK NumericStepper

MovieClip

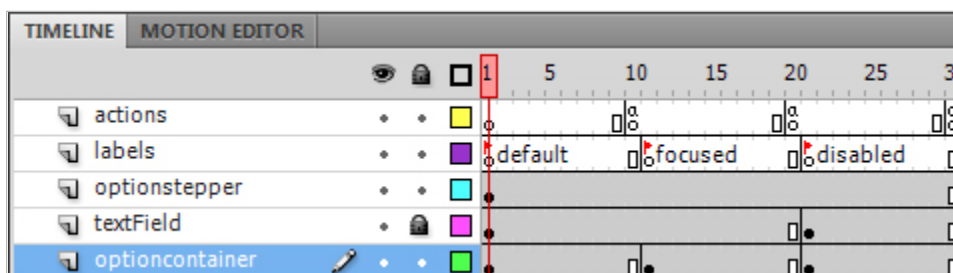
:

- **textField** TextField
- **nextBtn** CLIK Button dataProvider
- **prevBtn** CLIK Button dataProvider

2.4.2.3

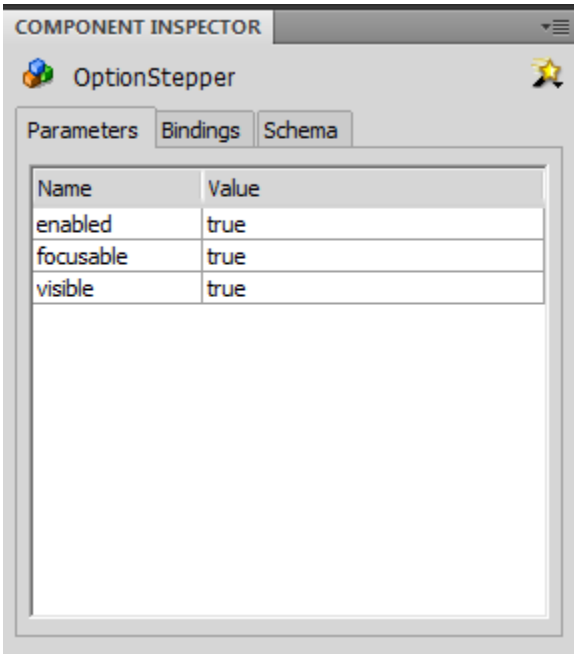
OptionStepper

- **default** enabled
- **focused** textField
- **disabled**



36: OptionStepper

2.4.2.4



OptionStepper		MovieClip	
Enabled	false,		
Focusable	/tab	focusable()	false
Visible	false		

2.4.2.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable*

OptionStepper

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false

ComponentEvent.STATE_CHANGE			
FocusHandlerEvent.FOCUS_IN			
FocusHandlerEvent.FOCUS_OUT			
IndexEvent.INDEX_CHANGE	OptionStepper		
	index:NumericStepper	int	-
	1(		1
	lastIndex:NumericStepper		int
	-1(		)
	1		
data:		dataProvider	AS3 Object

OptionStepper

```
myOS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
    trace("myOS.selectedItem: " + e.target.selectedItem);
    // Do something
}
```

2.4.3 ListItemRenderer



2: **ListItemRenderer.**

ListItemRenderer (scaleform.clik.controls.ListItemRenderer) CLIK Button
 ScrollingList TitleList DropdownMenu

2.4.3.1

ListltemRenderer Button
 ListltemRenderer
 ListltemRenderer

2.4.3.2

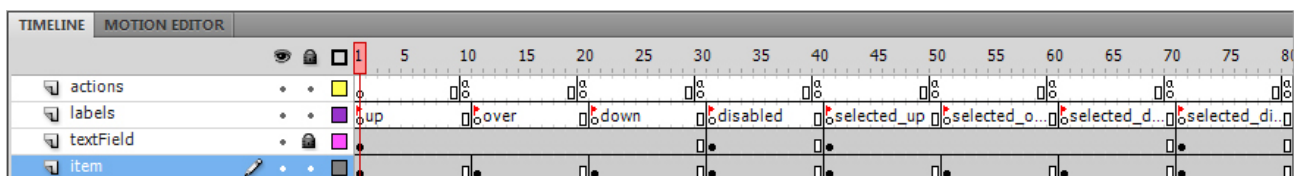
CLIK ListltemRenderer MovieClip :

- **textField** () TextField
- **focusIndicator**: () MovieClip MovieClip
 show hide

2.4.3.3

ListltemRenderer *selected*

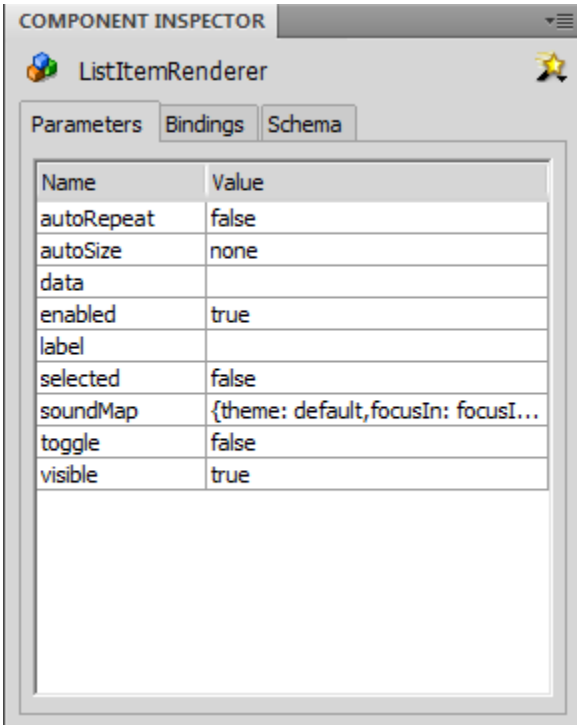
- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



38: ListltemRenderer

ListltemRenderer Button
 ListltemRenderer CLIK

2.4.3.4



ListItemRenderer

autoRepeat	ListItemRender	"click" ()
autoSize	ListItemRender ListItemRender autoSize=TextFieldAutoSize.NONE	autoSize
Data	ListItemRender TileList) ListItemRender	List() (ScrollingList /
Enabled	false,	
Label	ListItemRenderer	
selected	ListItemRender ListItemRender ListItemRender toggle true , , , toggle false, ActionScript	: toggle false,
Toggle	ListItemRender toggle	true, ListItemRender
Visible	false	

```

    Event( ) Event ,

    • type
    • target
    • currentTarget Event
    • eventPhase EventPhase.CAPTURING_PHASE
      EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
    • bubbles
    • cancelable

    ListItemRenderer ,

    , ListItemRenderer's , List(
  ) ListItemRenderer , ITEM_PRESS ITEM_CLICK
  ListEvents EventListener
(ListEvent.ITEM_CLICK) , ListItemRenderer EventListener

```

55

ButtonEvent.PRESS	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -
MouseEvent.DOUBLE_CLICK	<i>doubleClickEnabled</i> true mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -
ButtonEvent.DRAG_OVER	controllerIdx: () uint
ButtonEvent.DRAG_OUT	controllerIdx: () uint
ButtonEvent.RELEASE_OUTSIDE	controllerIdx: () uint

2.4.4 ScrollingList



39: **ScrollingList.**

ScrollingList (scaleform.clik.controls.ScrollingList)

ScrollIndicator

ScrollBar

dataProvider

```
scrollingList.dataProvider = new DataProvider(["item1", "item2", "item3", "item4"]);
```

crollingList

ListltemRenderer

FLA

2.4.4.1

ScrollBar

ScrollingList

ScrollingList

ScrollingList *wrapping*

wrapping

normal"

wrapping

wrap"

wrapping

stick"

(Page Up)

(Page down)

(Home)

(End)

ScrollBar

ScrollingList

ScrollaBar

Pad

CLIK

UI

2.4.4.2

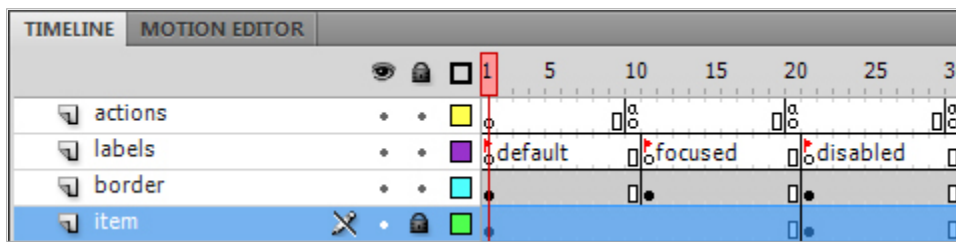
ScrollingList

ScrollingList

2.4.4.3

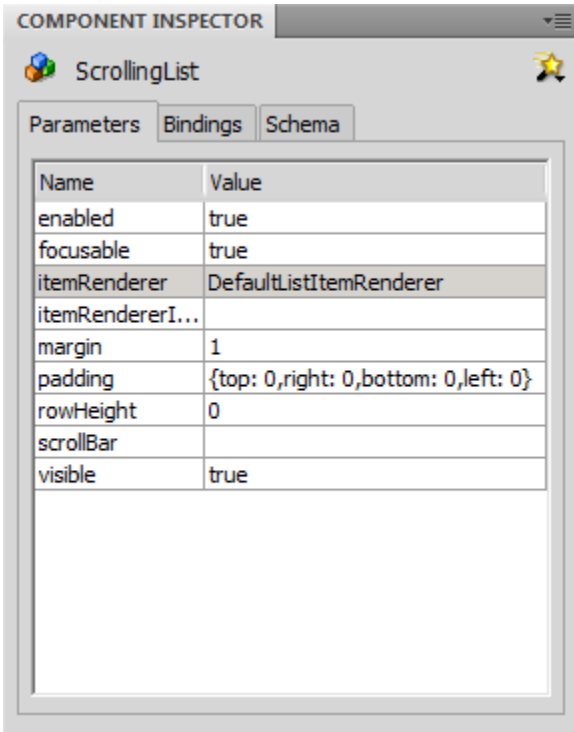
ScrollingList

- **default** enable
- **focused**
- **disabled**



40: ScrollingList

2.4.4.4



ScrollingList

MovieClip

Enabled	false,
Focusable	/ focusable() false tab
itemRenderer	ListItemRenderer itemRendererInstanceName (ListItemRenderers),
itemRendererInstanceName	ScrollingList r' r1' r2' r3'..... 1
Margin	rendererInstanceName
Padding	(Padding) itemRendererInstanceName (ListItemRenderer),

	ScrollBar()	
rowHeight	itemRendererInstanceName (ListItemsRenderer),	
Visible	false,	ScrollBar
	ListItemsRenderer	

2.4.4.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable*

ScrollingList

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
ListEvent.INDEX_CHANGE		
	• itemRenderer: IListItemRenderer • itemData: dataProvider ActionScript Object • index: int -1() 1 • controllerIdx: /	

ListEvent.ITEM_PRESS

- ## ListEvent.ITEM_CLICK

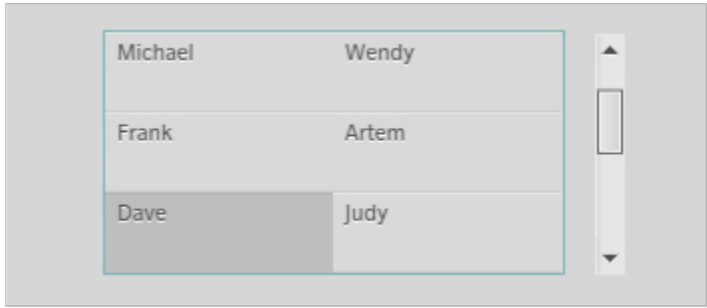
- ## ListEvent.ITEM_DOUBLE_CLICK

- ## ListEvent.ITEM_ROLL_OVER

	dataProvider		ActionScript
	Object		
• index:		int	-1(
	)		1
• controllerIdx:			/
	(		)
ListEvent.ITEM_ROLL_OUT			
• itemRenderer:			
	IListItemRenderer		
• itemData:			
	dataProvider		ActionScript
	Object		
• index:		int	-1(
	)		1
• controllerIdx:			/
	(		)

```
myList.addEventListener(ListEvent.ITEM_CLICK, onItemClick);
function onItemClick(e:ListEvent) {
    trace("ListItemRenderer Clicked: " + e.itemRenderer);
    // Do something
}
```

2.4.5 TileList



41: TileList.

TileList (scaleform.clik.controls.TileList)

ScrollingList

ScrollBar

TileList

ScrollIndicator
ScrollingList

dataProvider

```
tileList.dataProvider = new DataProvider(["item1", "item2", "item3",  
"item4", "item5"]);
```

FLA TileList ListItemRenderer

2.4.5.1

ScrollBar TileList

TileList

(Page Up) (Page down)
(Home) (End) ScrollBar

TileList ScrollBar

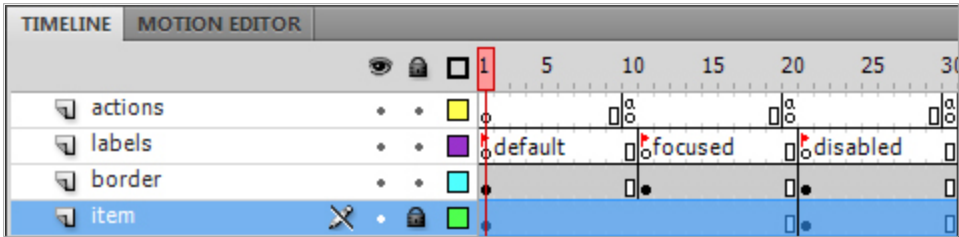
2.4.5.2

TileList TileList

2.4.5.3

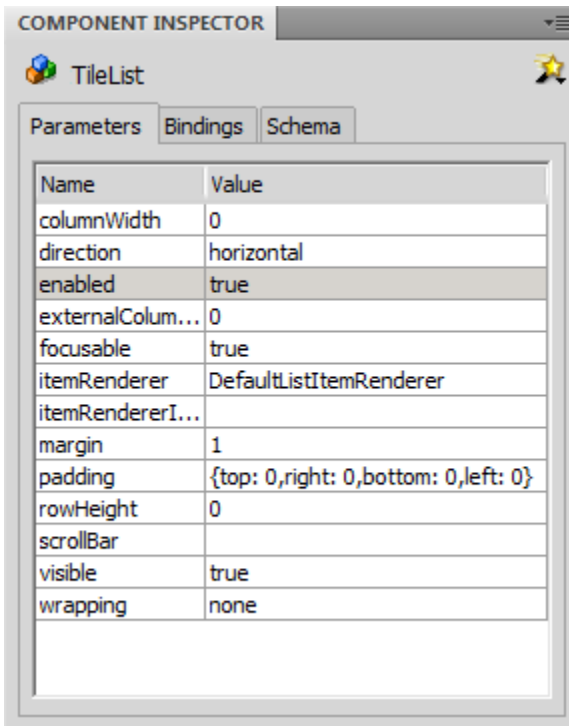
TileList

- **default** enable
- **focused**
- **disabled**



42: TileList

2.4.5.4



TileList

MovieClip

columnWidth	<i>rendererInstanceName</i>
Direction	
Enabled	false,
externalColumnCount	<i>rendererInstanceName</i> TileList
Focusable	/ focusable() false tab

itemRenderer	ListItemRenderer
	itemRendererInstanceName (
	ListItemRenderer),
itemRendererInstanceName	ScrollingList
	Stage
	'r',
	: 'r1', 'r2', 'r3', ... 1
Margin	itemRendererInstanceName (
	ListItemRenderer),
Padding	(Padding)
	itemRendererInstanceName (
	ListItemRenderer),
	ScrollBar()
rowHeight	itemRendererInstanceName (
	ListItemRenderer),
Visible	false, ScrollBar
	ListItemRenderer

2.4.5.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**
TileList

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		

FocusHandlerEvent.FOCUS_OUT

ListEvent.INDEX_CHANGE

- itemRenderer:
IListItemRenderer
- itemData:
dataProvider ActionScript Object
- index: int -1() 1
- controllerIdx: /
()

ListEvent.ITEM_PRESS

- itemRenderer:
IListItemRenderer
- itemData:
dataProvider ActionScript Object
- index: int -1() 1
- controllerIdx: /
()

ListEvent.ITEM_CLICK

- itemRenderer:
IListItemRenderer
- itemData:
dataProvider ActionScript Object
- index:
int -1() 1
- controllerIdx:
() /

ListEvent.ITEM_DOUBLE_CLICK

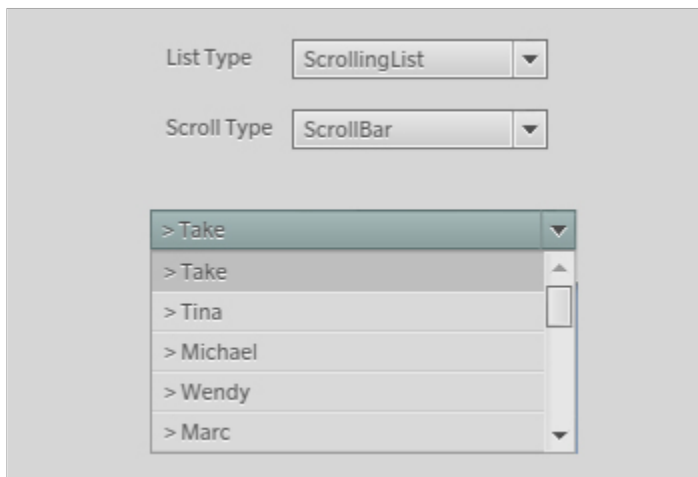
- itemRenderer:
IListItemRenderer
- itemData:

	dataProvider	int	-1(	ActionScript Object
	• index:	1	)	
	• controllerIdx:	/	(	
			)	
ListEvent.ITEM_ROLL_OVER	• itemRenderer:			
	IListItemRenderer			
	• itemData:			
	dataProvider			ActionScript Object
	• index:	int	-1(	
		1	)	
	• controllerIdx:	/	(	
			)	
ListEvent.ITEM_ROLL_OUT	• itemRenderer:			
	IListItemRenderer			
	• itemData:			
	dataProvider			ActionScript Object
	• index:	int	-1(	
		1	)	
	• controllerIdx:	/	(	
			)	

TileList

```
myList.addEventListener(FocusHandlerEvent.FOCUS_IN, onListFocused);
function onListFocused(e:FocusHandlerEvent) {
    trace("myList is now focused!");
    // Do something
}
```

2.4.6 DropdownMenu



43: **DropdownMenu.**

DropdownMenu (scaleform.clik.controls.DropdownMenu)

DropdownMenu

ScrollingList

TileList

ScrollBar

ScrollIndicator

dataProvider

DropdownMenu

dataProvider

dataProvider

```
dropdownMenu.dataProvider = new DataProvider(["item1", "item2", "item3", "item4"]);
```

DropdownMenu

ListItemRenderer

DropdownMenu

ListItemRenderer

FLA

DropdownMenu

ScrollBar

ScrollIndicator

DropdownMenu

2.4.6.1

DropdownMenu

(Enter)

ScrollingList

TileList

ScrollBar

DropdownMenu

2.4.6.2

DropDownMenu

MovieClip

- **textField** () TextField
 - **focusIndicator:** () MovieClip
- “show” “hide”

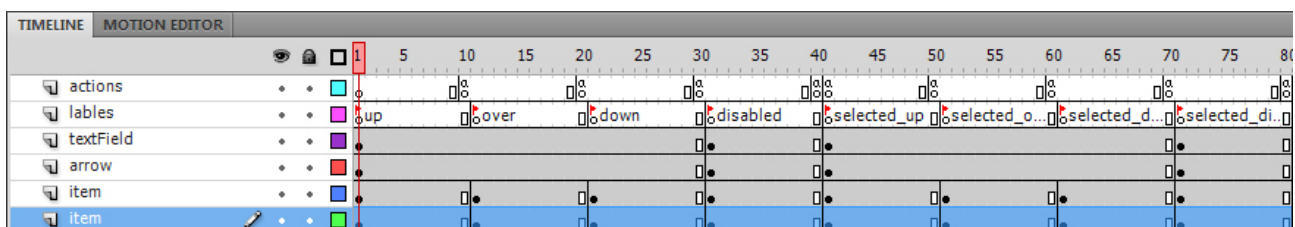
2.4.6.3

DropDownMenu

ToggleButton

CheckBox

- **up**
- **over**
- **down**
- **disabled**
- **selected_up**
- **selected_over**
- **selected_down**
- **selected_disabled**



44: DropDownMenu

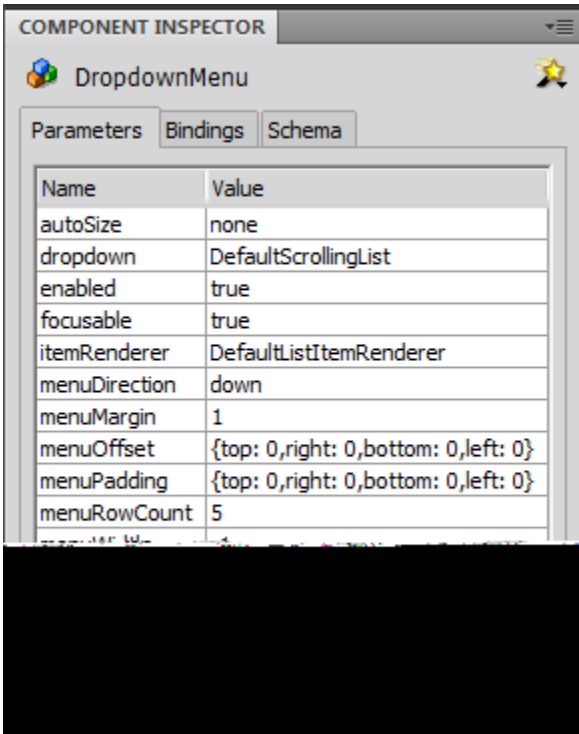
DropDownMenu

Button

DropDownMenu

[CLIK](#)

2.4.6.4



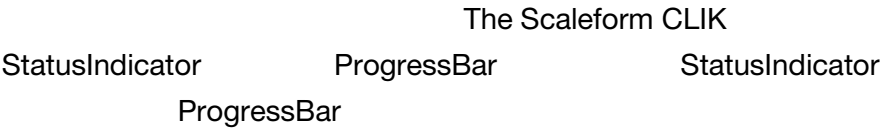
DropdownMenu

autoSize	DropdownMenu()		autoSize
	autoSize="none"		
Dropdown	ScrollingList	TileList	
	DropdownMenu		
Enabled	false,		
Focusable	,		false
menuDirection	"up"() "down"()		
menuMargin			
menuOffset			
menuPadding			
menuRowCount			
menuWidth	If set, this number will be enforced as the width of the drop down's list.		

	(	)
MouseEvent.ROLL_OVER	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
MouseEvent.ROLL_OUT	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
ButtonEvent.PRESS	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -	
MouseEvent.DOUBLE_CLICK	doubleClickEnabled true mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -	
ButtonEvent.DRAG_OVER	controllerIdx: (	

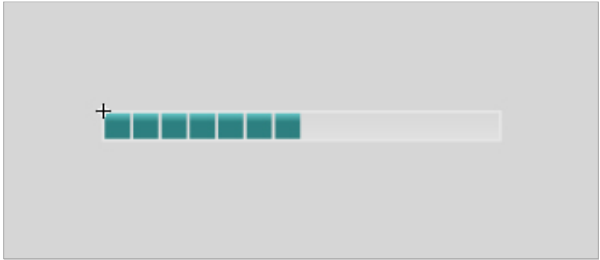
	) uint
ButtonEvent.DRAG_OUT	controllerIdx: (
	) uint
ButtonEvent.RELEASE_OUTSIDE	controllerIdx: (
	) uint

2.5



45: *Mercenaries 2* Faction

2.5.1 StatusIndicator



46: **StatusIndicator.**

StatusIndicator component (scaleform.clik.controls.StatusIndicator)

2.5.1.1

StatusIndicator

2.5.1.2

CLIK

StatusIndicator

MovieClip

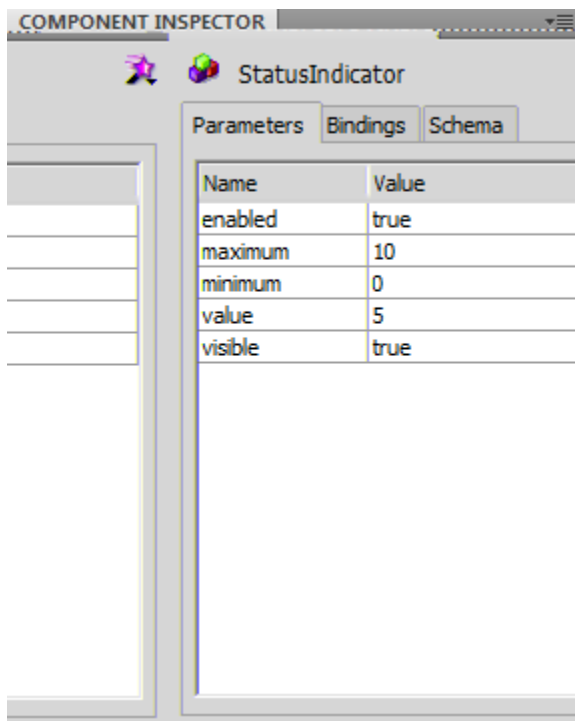
stop()

gotoAndStop()

2.5.1.3

StatusIndicator

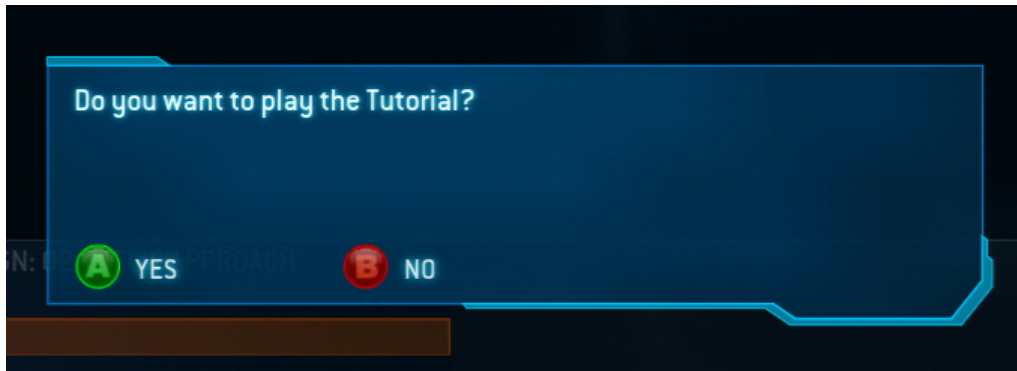
2.5.1.4



StatusIndicator

MovieClip

Visible	false	
Enabled	true	.



48: *Halo Wars* Window

2.6.1 Window

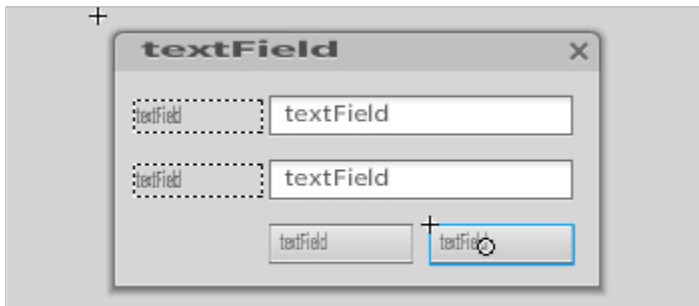


Figure 3: Sample unskinned Window.

CLIK Window (scaleform.clik.controls.Window), Alert()

PopUpManager : (PopUpManager.showModal())

2.6.1.1

Window

2.6.1.2

CLIK Window MovieClip :

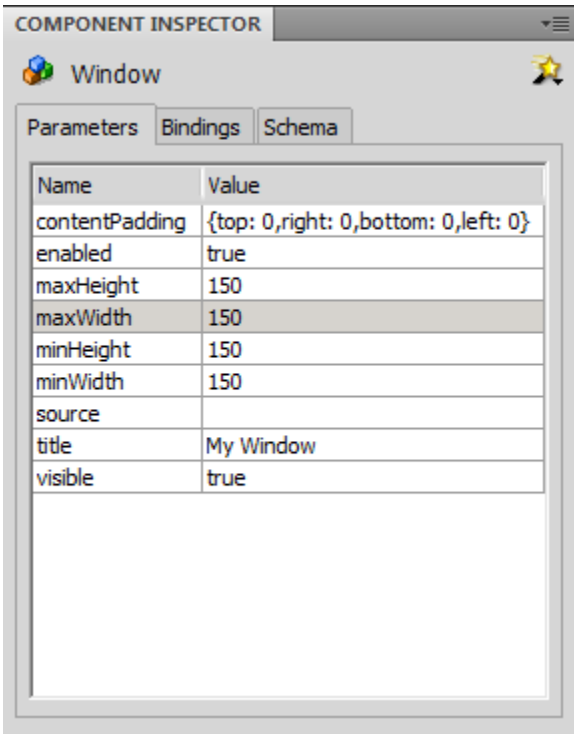
- **closeBtn** ()CLIK Button
- **titleBtn** ()CLIK Button

- **okBtn**)CLIK Button “Accept”() “OK”() ,
- **resizeBtn:**)CLIK Button , ,
- **background**)MovieClip ,
- **hit**)MovieClip hitArea

2.6.1.3

Window MovieClip

2.6.1.4



Dialog MovieClip

contentPadding		,
Enabled	true	.
maxHeight		
maxWidth		
minHeight		

minWidth	
Source	
Title	titleBtn ,
Visible	false

2.6.1.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE
EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

Window

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false

```
myWindow.addEventListener(ComponentEvent.HIDE, onWindowClosed);
function onWindowClosed(e:ComponentEvent) {
    // Do something.
}
```

3

Scaleform CLIK

3.1

Scaleform CLIK

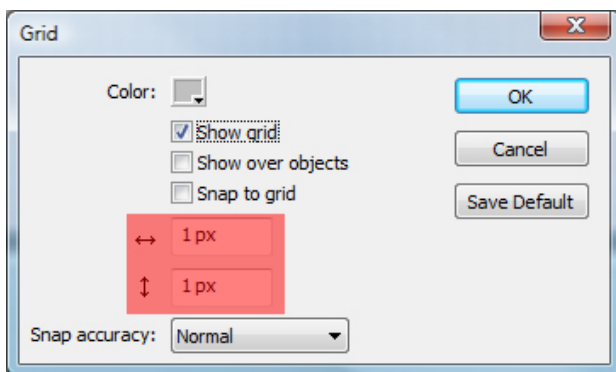
3.1.1

CLIK

Flash

1. Flash
- 2.
3. Flash
- 4.
- 5.
6. OK

1px

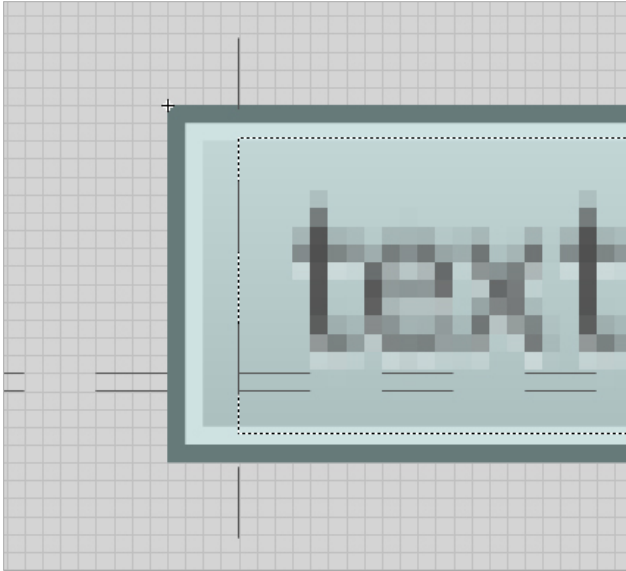


51: Grid

SWF

2

2



52:

3.1.2

Flash

Scaleform

Flash

Photoshop®

PNG

3.1.3

—

3.1.4

Flash

3.1.5

3.1.6 2

2 2

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

Flash

Scaleform CLIK

3.2.1

Flash

3.2.1.1

1

FLA

CLIK

FLA

1. *CLIK_Components fla*

2. FLA

FLA

Copy

3. FLA *Paste*

4. FLA *Rename*

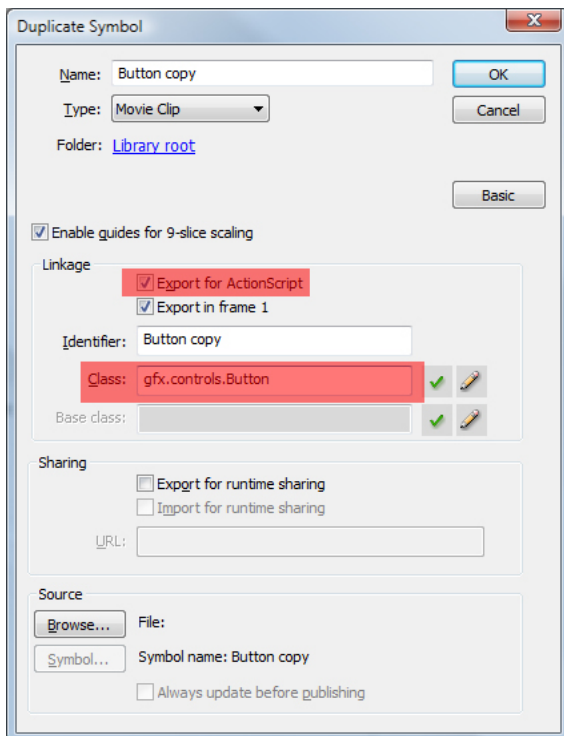
Button'

5. FLA *Properties*

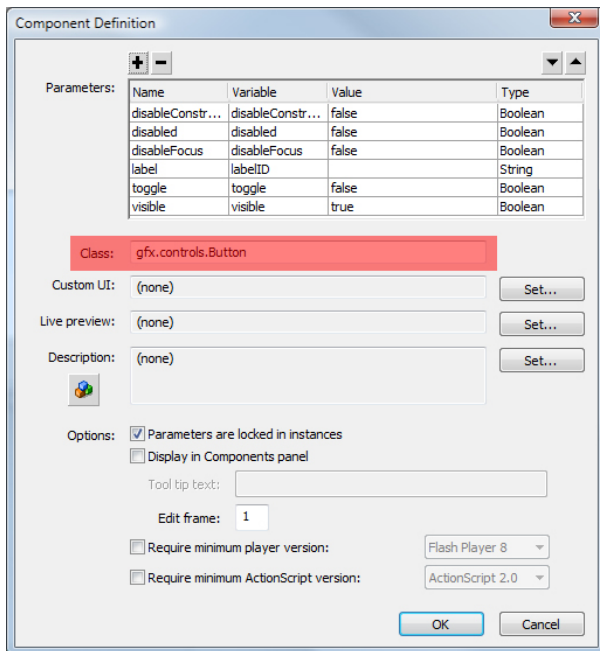
6. *Identifier* 3

6. *Paste*

1. *library* *Properties*
2. *Properties* *scaleform.clik.controlsButton* *Class*
(CTRL+C)
3. *Cancel*
4. *Duplicate*
5. *Duplicate Symbol* *Export* *ActionScript*
6. *Class* (CTRL+V) *textField*
7. *OK*
8. *library* *Component Definition*
9. *Class* *textField* (CTRL+V) *textField*
10. *OK*



53:



54:

3.3

Scaleform CLIK

CLIK

CLIK

- Flash
- Flash

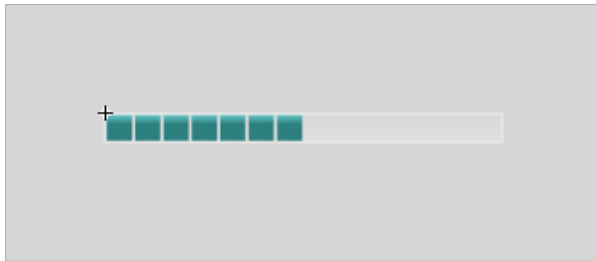
Flash

FLA

- Photoshop Illustrator® Flash

[CLIK](#)

3.3.1 StatusIndicator



55: StatusIndicator.

StatusIndicator

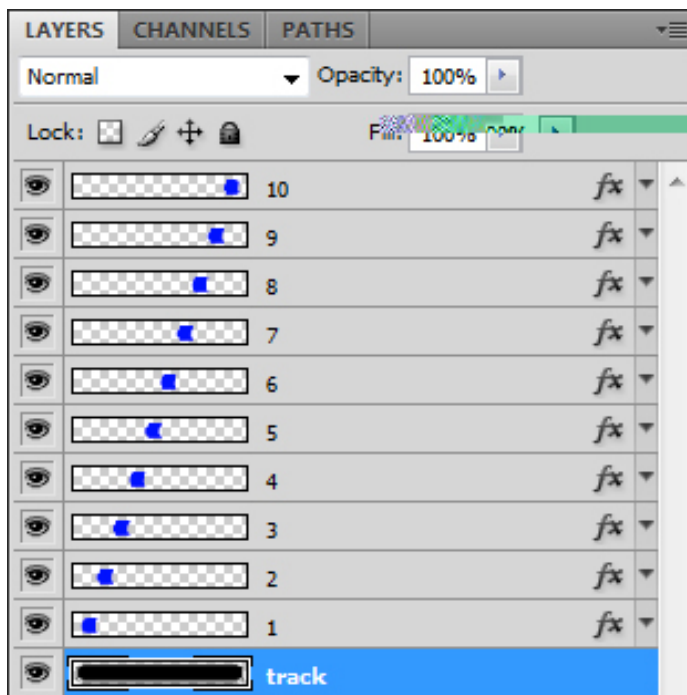
Indicator

Photoshop

indicator track Track

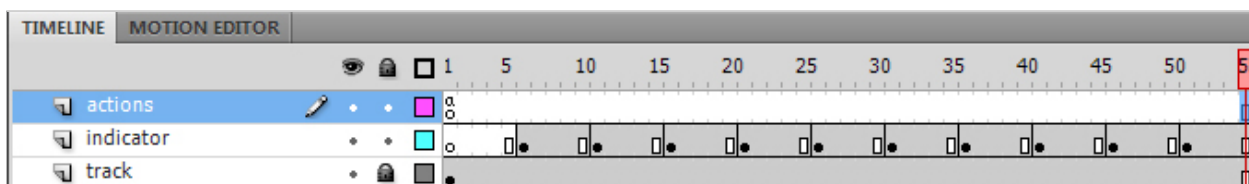


56: StatusIndicator PSD



57 Photoshop StatusIndicator PSD

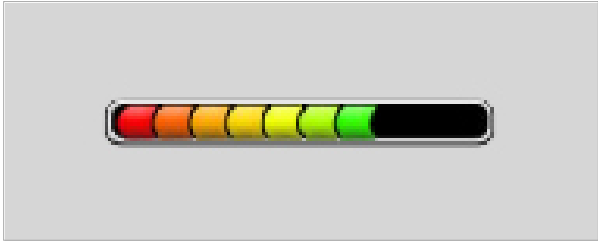
1. Photoshop StatusIndicator
2. PSD
3. Flash *File Import -> Import to Stage*
4. StatusIndicator PSD
5. *Import Convert layers to: Layers*
6. OK
7. Flash PSD
10. PSD 10 1
10. 10 1 10 layer 1
- 8.
- 9.
10. layer 1 6
11. Insert layer 1 11
12. layer 2 (Ctrl+X)
13. layer 1 11 (Ctrl+Shift+V) layer 1
14. 10 (layer 1)
- 5 layer 2 11
- layer 3 16 layer 4 21
- 10 PSD 51
15. 2-10
- 16.
- Remove Frames
- 55
17. indicator layer
18. track track
19. layer 1 indicator
20. indicator track actions layer track indicator



58:

21. StatusIndicator

- 22. 1 10
- 23.
- 24.



59: StatusIndicator

3.4

Scaleform CLIK

3.4.1

Scaleform SWF Scaleform
Flash UI

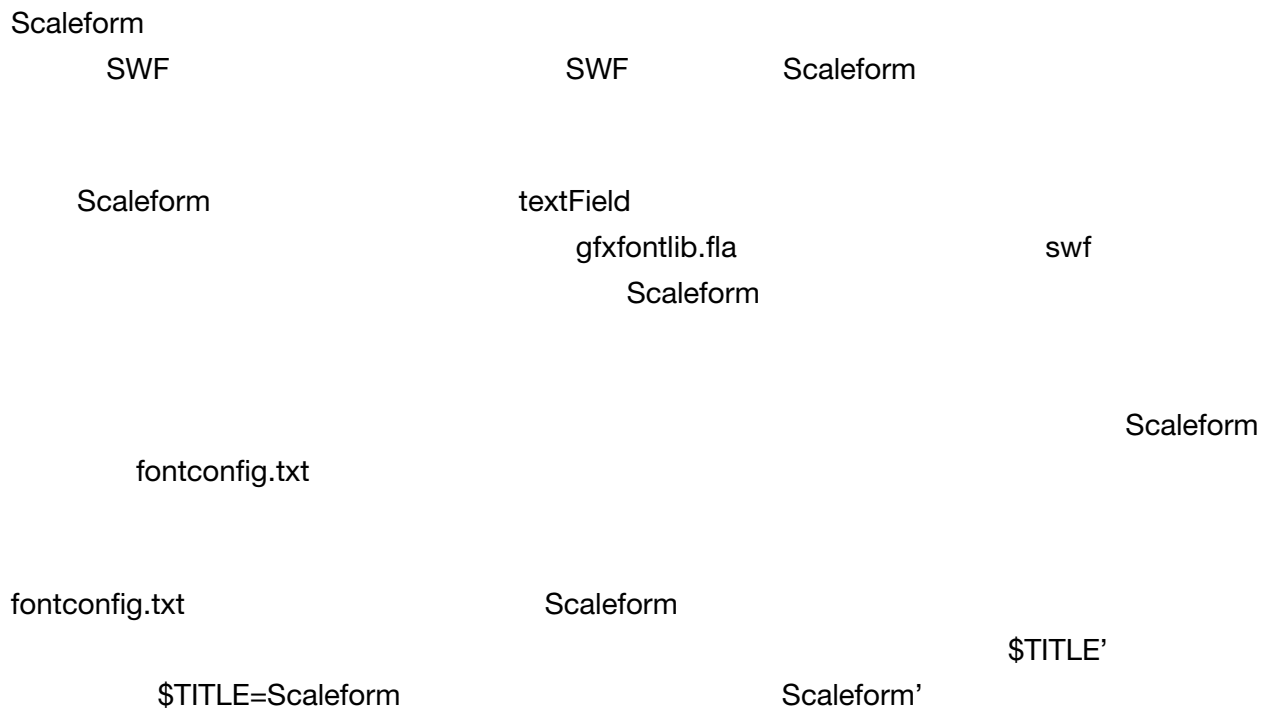
3.4.2

Flash Scaleform Scaleform
Slate Mobile textFidld Scaleform
Slate Mobile ASCII
Slate Mobile CJK
Scaleform textFields Scaleform 3.4.4
Scaleform Scaleform UI

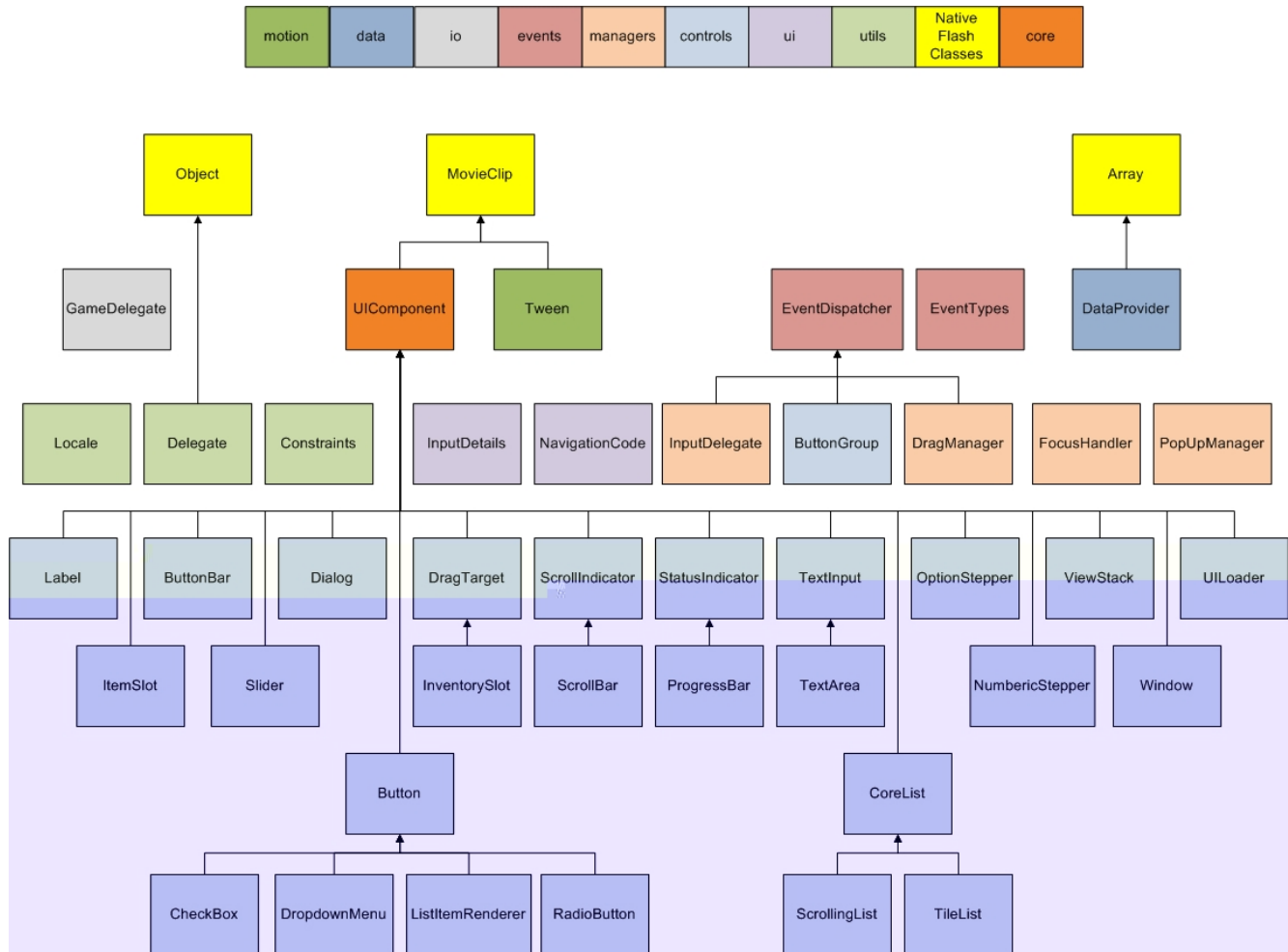
3.4.3 textField



3.4.4



Scaleform CLIK



60: Scaleform CLIK

4.1 UI *UIComponent*

Scaleform CLIK
(scaleform.clik.core.UIComponent).
UIComponent

UIComponent
Scaleform
CLIK

UIComponent
MovieClip

Flash 8

MovieClip

/

Flash
UIComponent

- **enabled;**
- **visible;**

- ***focused;***
- ***focusable;***
- ***width;***
- ***height;***
- ***scaleX;***
- ***scaleY;***
- ***displayFocus:*** true
- ***focusTarget:*** Settings this property to another MovieClip will cause that MovieClip to receive focus rather than this component if focus is ever set to this MovieClip. For more information see the [Focus Handling](#) section.

UIComponent

- ***configUI:***
- ***draw:***
- ***changeFocus:***
- ***scrollWheel:***

UIComponent EventDispatcher

4.1.1

- onLoad()
- MovieClip
- configUI()
- validateNow()
- draw()

4.2

Scaleform CLIK

4.2.1

ListItemRenderer RadioButton CheckBox CLIK Button
 Button Button (scaleform.clik.controls.Button)

- *up*
-
-
- *disabled*

Button
Button selected_” CLIK

- *up*
-
-
- *disabled*
- *selected_up*
-
- *selected_down*
- *selected_disabled*

CLIK

[CLIK](#)

CLIK getStatePrefixes()

```
protected function getStatePrefixes():Vector.<String> {
    return _selected ? statesSelected : statesDefault;
}
```

CLIK selected_” getStatePrefixes()

stateMap

CLIK

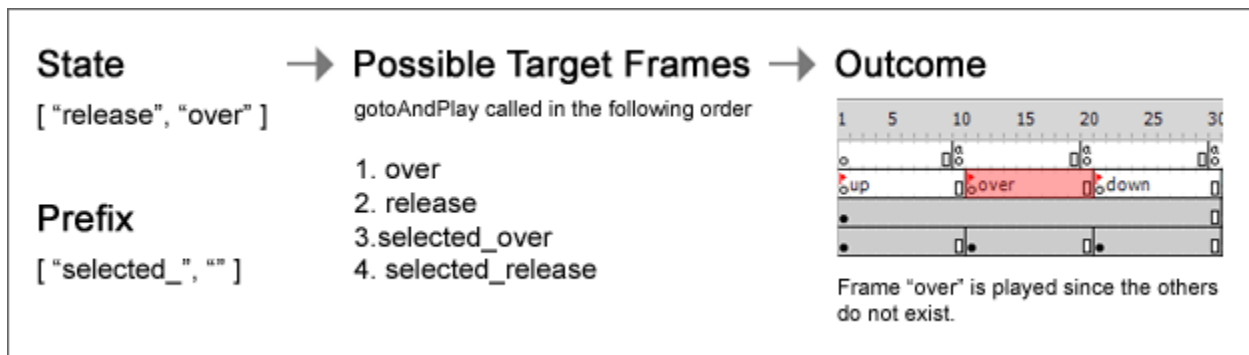
```
protected var _stateMap:Object = {
    up:["up"],
```

```

    over: ["over"],
    down: ["down"],
    release: ["release", "over"],
    out: ["out", "up"],
    disabled: ["disabled"],
    selecting: ["selecting", "over"],
    toggle: ["toggle", "up"],
    kb_selecting: ["kb_selecting", "up"],
    kb_release: ["kb_release", "out", "up"],
    kb_down: ["kb_down", "down"]
}

```

getStatePrefixes()



61:

4.2.2

CLIK

ScrollingList OptionStepper Slider TextArea

- *default*
- *focused*
- *disabled*

4.2.3

- *default*
- *disabled*

4.2.4

		StatusIndicator			ProgressBar
					50
0	10	5 50%	gotoAndStop()	25 50	50%
		updateValue()			

4.3

Scaleform CLIK

event model

ActionScript 3

CLIK

ActionScript 3

,

:http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

4.3.1

4.3.1.1

addEventListener()

removeEventListener()

,

CLIK

,

,

Event()

CLIK

scaleform.clik.events

,addEventListener()

,

:useCapture priority

useWeakReference

,

,

,
addEventListener()

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false,  
0, true);
```

addEventListener(),

[http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener\(\)](http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener())

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false,  
0, true);  
function onButtonClick (e:Event):void {  
    buttonInstance.removeEventListener(ButtonEvent.CLICK,  
onButtonClick, false);  
}
```

, buttonInstance ButtonEvent.CLICK
onButtonClick

, Event() CLIK

4.3.1.2

CLIK dispatchEvent()

```
dispatchEvent(new Event(Event.CHANGE));
```

4.4

ActionScript 3 ,

```
import flash.utils.getDefinitionByName;  
import scaleform.clik.controls.Button;
```

```
public var myButton:Button;
```

```
// Class
```

```

var classRef:Class = getDefinitionByName("ButtonLinkageID") as Class;
//
if (classRef != null) { myButton = new classRef () as Button; }
myButton.addEventListener(ButtonEvent.CLICK, onClick, false, 0, true);

    ,FLA
    , Class( ) "ButtonLinkageID", Base Class( )
scaleform.clik.controls.Button

.FLA
,
,
,

```

4.5

Scaleform CLIK

```

, CLIK FocusHandler
(scaleform.clik.managers.FocusHandler) FocusHandler

```

```

Scaleform
, ( )
,
ActionScript ActionScript
- myUIComponent.focused = 1; // 1 1
- stage.focus = myMovieClip; // MovieClip Stage
- FocusHandler.getInstance().setFocus(myMovieClip); // FocusHandler

```

4.5.1

```

InteractiveDisplayObject
Tab true

tabEnabled mouseEnabled false MovieClips Scaleform Flash
, CLIK tabEnabled

mouseEnabled

, tabEnabled mouseEnabled
;UIComponent.focusable , AS3 MovieClip.focusEnabled
false,UIComponent CLIK , focusable
tabEnabled mouseEnabled , tabEnabled
mouseEnabled

```

FocusHandler

displayFocus

true

Slider

UIComponent.changeFocus()

```
function changeFocus():void {
    track.displayFocus = _focused;
}
```

```

,
,
tab
,
,
,
tabEnabled
false

```

```
background.tabEnabled = false;
```

4.5.2

Flash Scaleform Selection

1. `tabEnabled = mouseEnabled = focusable = false`

OptionStepper

2. focusable = true

3.	displayFocus	true	changeFocus
----	--------------	------	-------------

4.6

Scaleform CLIK

Flash 8 MovieClip

MovieClip

CLIK

4.6.1

```
InputDelegate      class (scaleform.clik.managers.InputDelegate)
                    InputDetails      (scaleform.clik.ui.InputDetails)
                    InputDelegate.readInput()
InputDetails      InputDelegate
```

InputDetails

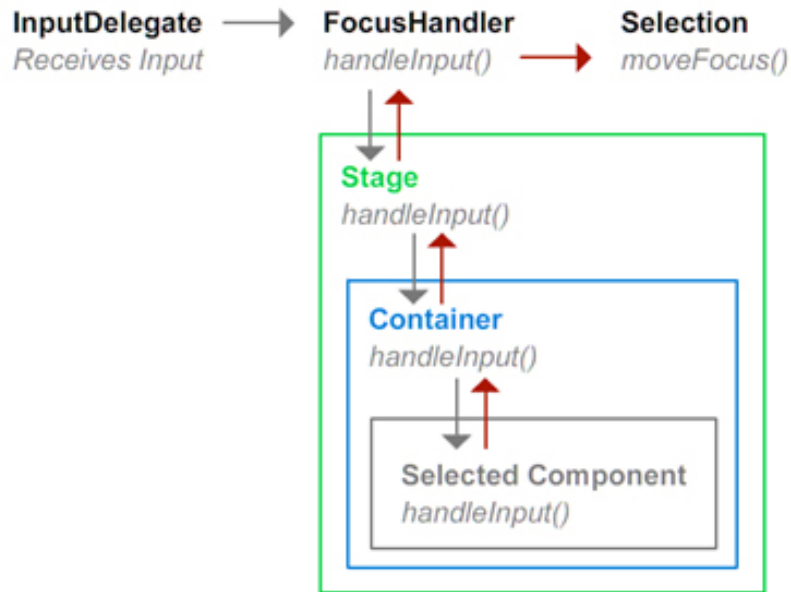
- key
-
-

InputValue.KEY_UP (“keyUp”) InputValue.KEY_DOWN (“keyDown”)

- navEquivalent(navigation equivalent) “as NavigationCode.UP (“up”),
NavigationCode.DOWN (“down”), NavigationCode.LEFT (“left”), NavigationCode.RIGHT
 (“right”), NavigationCode (scaleform.clik.ui.NavigationCode)

```
FocusHandler      InputDelegate      InputEvent      ,
InputEvent,      bubble
```

the handleInput()



62: handleInput()

```

InputEvent      ,handled      true
event.isDefaultPrevented(),

```

```

override public function handleInput(event:InputEvent):void {
    // Check whether the event has already been handled by another component.
    if (event.isDefaultPrevented()) { return; }
    var details:InputDetails = event.details;

    switch (details.navEquivalent) {
        case NavigationCode.ENTER:
            if (details.value == InputValue.KEY_DOWN) {
                // Do something.
                event.handled = true;
            }
            break;
        default:
            break;
    }
}

```

```

InputEvent      ActionScript 3      (bubble),      .bubble
,      ScrollingList,      InputEvent      ,
Enter      ,ListItemRenderer      ButtonEvent.CLICK      ,

```

```

override public function handleInput(event:InputEvent):void {
    if (event.isDefaultPrevented()) { return; }
}

```

```

    // Pass on to selected renderer first
    var renderer:IListItemRenderer = getRendererAt(_selectedIndex,
_scrollPosition);
    if (renderer != null) {
        // Since we are just passing on the event, it won't bubble, and should
properly stopPropagation.
        renderer.handleInput(event);
        if (event.handled) { return; }
    }
    . . .
}

```

InputDelegate.instance

FocusHandler

```

InputDelegate.instance.addEventListener(InputEvent.INPUT, handleInput);
function handleInput(e:InputEvent):void {
    var details:InputDetails = e.details;
    if (details.value == Key.TAB) { doSomething(); }
}

```

InputDelegate

W A S D

InputDelegate

4.6.2

Nintendo Wii™

CLIK

SWF

```

myButton.addEventListener(ButtonEvent.CLICK, onCursorClick);
function onCursorClick(event:ButtonEvent):void {
    trace(event.controllerIdx);
}

```

4.7

```
btnInstance.toggle = true;
btnInstance.selected = true;
btnInstance.data = obj;
```

63: CLIK

4.7.1

```

    public void draw() {
        if (isInvalid()) {
            invalidate();
            draw();
        }
    }

    public void invalidate() {
        invalidateData();
        invalidateSize();
    }

    public void validateNow() {
        draw();
    }
}

```

4.8

Scaleform CLIK

- 1.

- 2.

scale9grid

Flash scale9Grid , , container
Flash/Flex , ,

CLIK , Scaleform scale9Grid CLIK
,
, container () , ,

Flash IDE setSize()
Flash IDE
MovieClips LivePreviews Scaleform Player
CLIK

4.8.1 Scale9Grid

Scaleform Player Scale9Grid MovieClip
scale9grid Flash MovieClips scale9grid
Flash Player Scaleform
MovieClip scale9grid
Scale9grid

4.8.2

Constraints (scaleform.clik.utils.Constraints)
ScrollBar
Constraints
ScrollBar initialize()
draw()
draw()

```
override protected function initialize():void {
    super.initialize();

    // The upArrow doesn't need a constraints, since it sticks to top left.
    if (downArrow) {
        constraints.addElement("downArrow", downArrow, Constraints.BOTTOM);
    }
}
```

```

    }
    constraints.addElement("track", track, Constraints.TOP |
                          Constraints.BOTTOM);
}

override protected function preInitialize():void {
    constraints = new Constraints(this, ConstrainMode.REFLOW);
}

protected function draw():void {
    constraints.update(_width, _height);
}

```

4.9

dataProvider
Scaleform CLIK IDataProvider (scaleform.clik.interfaces.IDataProvider)
API
dataProvider
dataProvider
CoreList (ScrollingList, TileList) OptionStepper
DropDownMenu

4.9.1

DataProvider (scaleform.clik.data.DataProvider) initialize()
dataProvider ActionScript dataProvider
dataProvider
dataProvider IDataProvider

```

myComponent.dataProvider = new DataProvider([
    "data1",
    4.3,
    {label:"anObjectElement", value:6}]);

```

dataProvider

- *length:*

- *requestItemAt:* dataProvider
 OptionStepper
- *requestItemRange:* dataProvider
 ScrollingList
- *indexOf:*
- *invalidate:* dataProvider
 dataChange dataProvider

dataProvider

ActionScript 2

dataProvider ExternalInterface.call

4.10

Scaleform CLIK Tween (scaleform.clik.motion.Tween), Tween
 Scaleform CLIK Tween , ,fl.transitions.easing.*

tween, Tween(), () Sprite tween
 tween , Tween /

- *ease* Function()
- *easeParam* Object()
- *onComplete* Function Tween (closure)
- *onChange* Function Tween (/ (tick/frame))
- *data* Object Tween (Tween)
- *nextTween* Tween Tween Tween Tween ,
- Tween *pause=false*
- *frameBased* Boolean ,
- *delay* Number tween x
- *fastTransform* Boolean
- *paused* Boolean Tween

Tween Tween MovieClip Tween Tween
Tween MovieClip
 Tweens

4.10.1

Tween MovieClip MovieClip tween

 tween , , onComplete

Tween Tween Tween ,

 Tween

```
import fl.transitions.easing.*;
import scaleform.clik.motion.Tween;

// Perform a 1 second tween from the current horizontal position and
// alpha values to the ones specified
var myTween:Tween = new Tween(300,
                               myMovieClip,
                               { x:100, y:100, alpha:0 },
                               { ease:Strong.easeOut,
                                 onComplete:onCompleteCallback} );

function onCompleteCallback(t:Tween):void { // Do something. }
```

4.11 佈局框架

CLIK

,CLIK UI, CLIK

:Scaleform v4.0.14 CLIK AS3 , API

CLIK AS3 , Scaleform

- CLIK
- **scaleform.clik.layout.Layout** Sprite , Sprite
 .layoutData
 - **scaleform.clik.layout.LayoutData** Sprite
- ,

CLIK

1. Sprite/MovieClip Stage DisplayObjectContainer

```

2.      LayoutData          Sprite/MovieClip      .layoutData
3.          Sprite/MovieClip      LayoutData          Layout
4.          1      Sprites/MovieClips      DisplayObjectContainer

      ,      Layout(      )      ,      Layout      Sprite
MovieClip,      Sprite      MovieClip      (x y      ),
      .layoutData

Scaleform\Resources\AS3\CLIK\demos\      SDK
Layout_Main      CLIK      ,      Scaleform SDK      Layout_Main
      MovieClip      Layout_FullScreen_Demo
      FxPlayer      ,      Stage      Layout_FullScreen_Demo
      Layout

```

4.11.1

```

(Layout)      ,      /
      Sprite(      )      ,
      ,
(myWindow.myMovieClip      myWindow.layoutInstance)      layoutInstance      myMovieClip

      Sprite,      Sprite      LayoutData      .layoutData
      ,      ,      .identifier      ,      Sprite
LayoutData      .layoutIdentifier      Stage      ,
      ,      .layoutData

      ,Layout
      .layoutData,Layout
Sprite/MovieClip      Sprite/MovieClip      Stage
      ,

```

public function reset():void

```

      Sprite      LayoutData      Sprite      /
      Sprite,      Layout

```

public function resortManagedSprites():void

```

      layoutData.layoutIndex      Sprite      ,
      layoutIndex      ,

```

```

        Layout Flash (Symbol) ,
Stage Layout Symbol( 0 ),
    Layout Layout .rect ,
Layout
    ,Layout , Sprite/MovieClip
LayoutData

```

4.11.1.1

Rect	“ “ x y
	Layout != 0(Symbol, Stage), Layout MovieClip x y Layout == 0(addChild() Symbol), Layout x y .rect Rectangle()
tiedToStageSize	true – Layout Stage , false
tiedToParent	true – Layout , false
Hidden	true - Layout, false Layout visible = false

4.11.2 LayoutData

Sprite scaleform.clik.layout.Layout

4.11.2. 1

alignH	Sprite LayoutMode.ALIGN_NONE LayoutMode.ALIGN_LEFT LayoutMode.ALIGN_RIGHT
alignV	Sprite LayoutMode.ALIGN_NONE LayoutMode.TOP LayoutMode.BOTTOM.property TextFieldAutoSize.NONE
offsetH	Layout relativeToH Sprite (-1),Layout Stage Layout/Sprite ()
offsetV	Layout relativeToH Sprite (-1),Layout Stage Layout/Sprite

	(		)
offsetHashH	(Hash Table),	Layout	Stage (Layout.tiedToStageSize == true), , Layout Stage , 4:3,"LayoutData.offsetHashH[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px
offsetHashV	(Hash Table),	Layout	Stage (Layout.tiedToStageSize == true), , Layout Stage , 4:3,"LayoutData.offsetHashV[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px
relativeToH	Sprite	Sprite	null,Sprite
relativeToV	此 Sprite 應與其垂直相關的 Sprite 的實例名稱。假如保留為 null,Sprite 就會相對於 Layout 而對齊。		
layoutIndex	Layout	Sprite	Sprite
	-	relativeToH	relativeToV
	Sprite	Layout	layoutIndex (-1), Sprite
layoutIdentifier	LayoutData	Layout	(
	DisplayObjectContainer	Layout)	
	(null),	Layout	(
	)	Layout	

4.12

Scaleform CLIK PopUpManager (scaleform.clik.managers.PopUpManager)

4.12.1

PopUpManager ID

initProperties , PopUp

 x y relativeTo ,Sprite ,

 Sprite

```
import scaleform.clik.managers.PopUpManager;
```

```

    PopUpManager.createPopUp("PopupLinkageID", {alpha:.5}, 100, 100
myMovieClip);

```

```

Scaleform      InteractiveObjectEx.setTopmostLevel()
                                     root
                                     SWF
                                     topmostLevel

InteractiveObjectEx.setTopmostLevel()
                                     Z
                                     topmostLevel

true

```

4.13

```

DragManager      (scaleform.clik.managers.DragManager)

,
DragManager.handleStartDragEvent()
    DragManager      ,
DragManager.handleEndDragEvent()
    DragManager      ,
    DragEvent.DRAG_STATE
    MouseEvent.MOUSE_UP
,

```

4.13.1



64: DragDemo

DragManager IDragSlot
DragManager

IDragSlot
DragManager

Scaleform IDragSlot
DragSlot UIComponent
DragSlot
DragSlot

DragSlot scaleform.clik.controls.DragSlot)
CLIK DragManager
/ /Bitmap DragSlot
DragSlot IDragSlot

DragSlot DragSlot
DragSlot DragManager dragBegin dragEnd
DragSlot
DragSlot UI

5

Scaleform CLIK

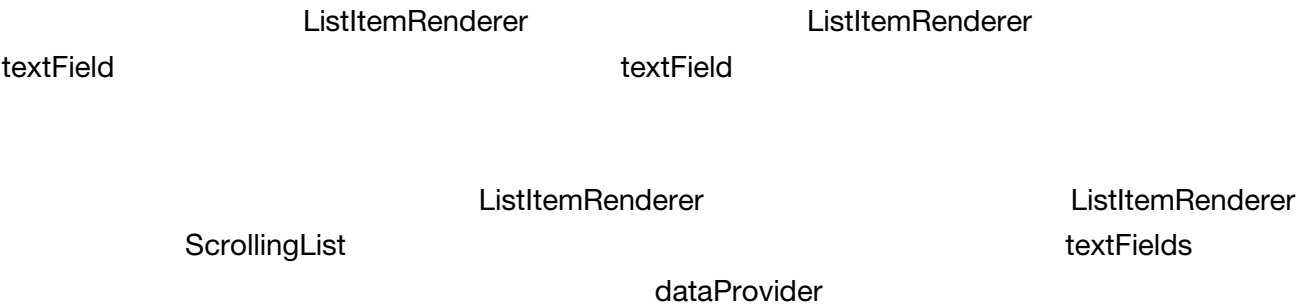
5.1

Scaleform CLIK

5.1.1 包含兩個 `textField` 的 `Listltem Renderer`



65: ScrollingList



```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},  
                     {fname: "Roger",   lname: "Federer"},  
                     {fname: "Michael", lname: "Schumacher"},  
                     {fname: "Tiger",   lname: "Woods"},  
                     {fname: "Babe",    lname: "Ruth"},  
                     {fname: "Wayne",   lname: "Gretzky"},  
                     {fname: "Usain",   lname: "Bolt"}];
```

textField fname lname

```

ListIteMRenderer      textField      textField
      ListIteMRenderer      MyIteMRenderer
ListIteMRenderer      MyIteMRenderer.as
      FLA

```

```
import scaleform.clik.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

    public var textField1:TextField;
    public var textField2:TextField;

    public function MyItemRenderer() { super(); }

    override public function setData(data:Object):void {
        this.data = data;
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }

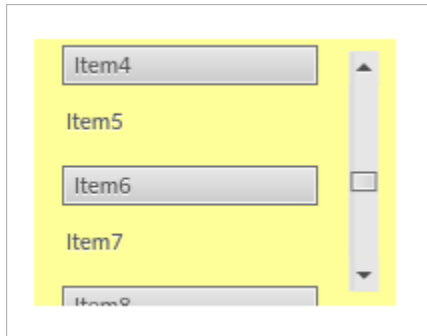
    override protected function updateAfterStateChange():void {
        super.updateAfterStateChange();
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }
}
```

```

graph TD
    L1[ListItemRenderer] -- setData --> L2[updateAfterStateChange]
    L1 -- setData --> L3[ListItemRenderer]
    L2 -- (ScrollingList TileList ) --> L4[textField]
    L2 -- (ScrollingList TileList ) --> L5[textField]
    L4 -- MyItemRenderer --> L6[updateAfterStateChange]
    L5 -- MyItemRenderer --> L6
    L6 -- textField --> L7[ListItemRenderer]
    L7 -- dataProvider --> L8[ListItemRenderer]
    L8 -- textField --> L9[textField1']
    L8 -- textField --> L10['textField2']
    L9 -- MyItemRenderer --> L11[MyItemRenderer']
    L10 -- MyItemRenderer --> L11
    L11 -- FLA --> L12[FLA]
    L11 -- dataProvider --> L13[dataProvider]
    L11 -- fname --> L14[fname]
    L12 -- lname --> L15[lname]

```

5.1.2 圖元滾動視圖



66:

CLIK

CLIK

ScrollBar

- **actions:** ActionScript
- **scrollbar:** CLIK ScrollBar sb
- **mask** MovieClip
- **content**
- **background**

content' y 0

ActionScript

code

```
// 133 is the view size (height of the mask)
sb.setScrollProperties(1, 0, content.height - 133);
sb.position = 0;

sb.addEventListener(Event.SCROLL, onScroll, false, 0, true);
function onScroll(e:Event) {
    content.y = -sb.position;
}
```

setScrollProperties

6 常見問題解答

1.

```
1152: A conflict exists with inherited definition scaleform.clik...
in namespace public.          public
scaleform.clik...
```

, .FLA
"Automatically Declare Stage Instances"(ActionScript .FLA .FLA
)
CLIK CLIK , Stage ,
ActionScript

File -> Publish Settings -> "Flash" tab -> Actionsript 3.0 Settings
"Automatically declare stage instances", .FLA "Automatically Declare Stage
Instances" , Flash Studio , .FLA

2. .FLA CLIK Stage
(inspectable) ,

```
1046: Type was not found or was not a compile-time constant:
...
```

-
- ActionScript
- ActionScript , -
ActionScript ,myButton:Button (scaleform.clik.controls.),
myButton:MyButtonSymbol, ,MyButtonSymbol , myButton

3. , .SWF ,

```
Symbol 'MySymbol', Layer 'actions', Frame 10, Line 1 -- 1024: Overriding a function
that is not marked for override.
```

```
Symbol 'MySymbol', Layer 'actions', Frame 20, Line 1 -- 1024: Overriding a function
that is not marked for override.
```

```
Symbol 'MySymbol', Layer 'actions', Frame 30, Line 1 -- 1024: Overriding a function
that is not marked for override.
```

```
“scaleform.clik.controls.Button” , “Button”
```

7 潛在缺陷

```
getDefinitionByName()
```

8 CLIK AS3 vs. CLIK AS2

This section outlines major differences, mostly programming related, between CLIK AS2 and CLIK AS3.

1. CLIK AS3 is designed to work in Flash Player.
2. Rather than using a custom EventDispatcher, CLIK AS3 uses ActionScript 3's native event system.
3. The invalidation system has been redesigned to help avoid unnecessary updates in draw(). Each component now stores an "invalid" table that tracks which aspects of the component are currently invalid. Default invalidation types are defined in `scaleform.clik.constants.InvalidationType`. Aspects can be marked as invalid using `invalidate()`, `invalidate[InvalidationType]()`, or manually setting the property to true within the table.. Logic within `draw()` should be wrapped within an appropriate `isInvalid()` check to ensure that unnecessary updates to components are avoided.
4. `invalidate()` no longer uses a 1ms delay; instead, it uses the next stage invalidation (`Event.RENDER`) or the next `Event.ENTER_FRAME` for the component and then calls `validateNow()`.
5. Tween's syntax is no longer `MovieClip.TweenTo`; instead, `var t:Tween = new Tween();`
6. `handleInput()` is now tied to the native event system. This means that `handleInput()` now accepts one parameter of type `InputEvent` and the event will bubble up from the component that dispatched it rather than being passed down from `FocusHandler`.
7. `handleInput()` no longer returns a Boolean; instead, it should set `event.handled = true`; if the event was handled.
8. `UIComponent` has a new property, `focusable`, which can be used with the CLIK `FocusManager` to prevent focus from reaching a component. This is more or less a replacement for AS2's `MovieClip.focusEnabled` property.
9. `enableInitCallback` is no longer an inspectable of `UIComponent`. Instead, it should be set within a class's constructor or `preInitialize()` method.
10. `DataProvider` now requires that you provide the target array to `DataProvider`'s constructor (`myComponent.dataProvider = new DataProvider(myArray);`), rather than simply setting `myComponent.dataProvider = myArray`;
11. `ButtonBar` will now only create Buttons that fit within its own size. If your `ButtonBar`'s size is 200px wide and each Button is 150px wide, only one Button will be displayed. This allows users to resize their `ButtonBar` and have the Buttons within be added and removed dynamically.
12. `UILoader` has been removed. This functionality is now handled by the Flash `MovieClip/Sprite` and `Loader` classes.
13. `UIComponent.SoundMap`, introduced in CLIK 3.3, has been removed for CLIK AS3.

